Tactical choice in repeated protests

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Abstract

Why do some protest movements escalate from peaceful demonstrations to violent confrontation while others never change tactics? I develop a dynamic model in which an organizer chooses between tactics that differ in how they convert participation into political power. Failed protests reveal information about regime strength, reducing subsequent mobilization. Optimal tactical sequencing reflects not only how powerful each tactic is today, but how valuable it may be in a lower-mobilization environment tomorrow. I show that movements preserve tactics that remain effective after demobilization, generating path-dependent patterns of tactical persistence or learning-motivating switching. Extending the model to repression, I show that state intervention can redirect movements across tactics and, by changing the informational content of failures, partly or even fully undermine the benefits of repression. As a technical contribution, I introduce a cost-signals structure which resolves equilibrium multiplicity in a dynamic global game with public learning.

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1 Introduction

What determines how protest movements choose and change tactics over time? Some movements undergo striking transformations in their main mode of contestation: The African National Congress turned from civil disobedience to violent sabotage and insurgency with the creation of Umkhonto we Sizwe, and the initially peaceful mass demonstrations during the Arab Spring gave way in some cases to violent clashes and armed confrontation. Other movements remain remarkably consistent in their tactical approaches: Poland's Solidarity movement relied overwhelmingly on nonviolent, mass-based contention throughout the 1980s, while a century earlier, the January Uprising of 1863 took the form of armed insurrection from the start. What explains these divergent trajectories, and how do movement leaders decide whether to persist with or depart from their initial tactics?

Canonical theoretical accounts emphasize how nascent protest movements grow by exposing latent sympathies, lowering barriers to participation, and generating mobilization cascades (Kuran 1991; Lohmann 1994). Yet the same informational feedback loop that helps launch a protest can work in reverse as failed protest episodes accumulate. When a movement repeatedly falls short of its objectives, citizens become disheartened: they have discovered that the regime is stronger, and the movement's prospects worse, than they previously believed. Failure therefore erodes confidence in subsequent attempts and, with it, the participation on which many forms of contention depend. Thus, while existing accounts tell us a great deal about how information causes movements to grow or peter out, they have less to say about how movements *change what they do* as public confidence deteriorates.

This paper argues that tactical choice allows movement leaders to navigate a dynamic trade-off between immediate political power and the consequences of failure for future mobilization. Because leaders are strategic actors who anticipate these consequences, the mere prospect of future demobilization can shape their choices from the outset. The key is that different tactics convert participation into political leverage in different ways. Mass-based tactics such as marches, occupations, and popular uprisings generate leverage through numbers: by demonstrating the breadth of opposition and paralyzing the normal functioning of society, they can compel policy concessions or regime change. By contrast, tactics such as sabotage or insurgency seek to achieve change by exerting strong coercive pressure with relatively few participants. Such tactics typically rely on resources, capital, or arms to damage infrastructure, disrupt economic activity, or threaten state assets.

These differences mean that the most powerful tactic today need not be the best tactic to deploy first. Leaders anticipate how failures will affect turnout, and thereby alter the relative effectiveness of tactics. But they can also *influence* that change: a failure by a more powerful protest provides stronger evidence of regime strength, potentially discouraging future participation more sharply. Tactical choice thus allows leaders not only to adapt to declining mobilization, but also to influence how much mobilization declines in the first place. In short, leaders deliberately sequence tactics to both capitalize on expected turnout today, and strategically position their movement for subsequent attempts.

I formalize these ideas in a two-period model in which an organizer chooses between two stylized protest technologies, which I call peace and violence. These labels distinguish tactics by how their leverage varies with participation, rather than whether they literally involve violence. *Peace* represents a participation-intensive technology, generating little leverage when turnout is low but scaling strongly with participation. *Violence*, by contrast, generates substantial leverage even with limited participation, but benefits less from additional turnout. Thus, even a literally violent mass revolution may be better captured by the former technology if its effectiveness depends primarily on the mobilization of large numbers. After the organizer chooses a tactic, citizens decide whether to participate, and protest succeeds if the power generated exceeds the unknown strength of the regime. If protest in the first period fails, citizens infer that the regime is stronger than previously believed, reducing their willingness to participate, and the game is played again in period 2.

My first set of results shows that even when the violent tactic is extremely effective, it may not be deployed it immediately – or even at all. In fact, organizers may deliberately preserve violence for later, and following a sufficiently *severe* failure, continue to use peace in the second period.

The first reason is that organizers need not deploy their highest-leverage tactic immediately. Because tactics differ in their sensitivity to participation, their relative effectiveness changes as mobilization declines. Violent tactics remain effective with relatively few participants remain effective even in pessimistic, low-mobilization environments. By contrast, participation-intensive "people power" tactics are *perishable*: their effectiveness depends on broad mobilization, which can dissipate when setbacks undermine public confidence. This creates a "use-it-or-lose-it" logic: even when violence offers greater immediate leverage, an organizer may begin peacefully, preserving violence as insurance against future demobilization. This incentive is strongest when turnout is particularly sensitive to setbacks: the more sharply failures reduce participation, the more valuable

it becomes to deploy participation-intensive tactics early, when they are likeliest to be most effective.

The second reason is that larger first-period failures make the peaceful tactic more appealing in the second period. Failure has two opposing effects on subsequent tactical choice. It reduces expected turnout, increasing the appeal of violent tactics, which can generate pressure with few participants. However, it also provides evidence that the regime is strong, raising the amount of leverage required for any subsequent challenge to succeed. After smaller failures, the first force dominates. However, after sufficiently severe failures, the second force outweighs it: Violence may remain relatively effective at low turnout, yet still lack the upside needed to overcome a strong regime. The organizer therefore "gambles" on peace, the only tactic which can generate enough pressure to make victory possible. The result is a *selection-based* account of why not all failed protests result in subsequent violence: the failures that most sharply depress mobilization may also make participation-intensive, high-upside tactics comparatively appealing for the future.

These results show that different tactical trajectories need not reflect fundamentally different strategic logics; they can emerge from the same dynamic problem. The model exhibits parameter regions in which movements remain peaceful throughout; others in which movements begin peacefully and turn to violence after weak failures; and still others in which movements begin violently but switch to peace after sufficiently severe failures. This generates predictions about how changes in the political and physical environment drive tactical sequencing decisions. Urbanization and urban redevelopment, for example, can reduce the leverage available to small groups while making street fighting, occupations, or control of public space more dependent on large crowds, weakening violence as a post-demobilization fallback even while strengthening its appeal during periods of broad mobilization.

In an extension, I study how repression interacts with the baseline dynamics. Policing, restrictions on public space, and investments in security alter the tactical environment – but their efficacy is limited by two factors. The first is simple substitution logic: weakening one tactic can redirect contention rather than suppress it. As a result, the direction of tactical change induced by repression can run opposite to the direction of change induced by learning alone in the baseline model.

The second limit is informational. Repression may help the regime survive an initial protest, but it also changes what citizens learn from failure. Failure under little repres-

sion is strong evidence that the regime itself is difficult to defeat; failure under heavy repression can instead be attributed partly to the intervention, leaving citizens less pessimistic. Repression therefore trades off immediate protection against the informational consequences of survival. When mobilization is highly responsive to beliefs, this can be self-defeating: by making failure less discouraging, repression preserves optimism and future turnout, leaving the movement better positioned to re-mobilize.

The paper proceeds as follows. Section 2 discusses related literature, and Section 3 presents the baseline model. Section 4 abstracts away from tactical choice to isolate the relationship between learning from failures and mobilization. Section 5 analyzes tactical choice. Section 6 extends the model to repression, and Section 7 concludes.

2 Literature

The paper builds on the classic literature on repertoires of contention, which emphasizes that movements draw upon historically and organizationally conditioned sets of contentious tactics (Tilly 1978). Within these repertoires, movements innovate and adapt their tactics in response to changing political circumstances (McAdam 1983). My contribution is to examine the strategic logic of *sequencing* tactics within a given repertoire. Rather than explaining how new tactics emerge or become available, I study how organizers choose among existing tactics as previous episodes of contention change the conditions under which they can be effectively deployed.

The model also contributes to a growing theoretical literature on dynamic coordination under incomplete information (Dasgupta 2007; Dasgupta, Steiner and Stewart 2012; Gieczewski and Koçak 2024; Mathevet and Steiner 2013; Shadmehr 2019), and in particular the literature that uses the global games approach originated by Morris and Shin (1998, 2003). The most direct antecedent is Angeletos, Hellwig and Pavan (2007) who model dynamic global games where regime survival conveys public bad news about regime strength. Little (2017) studies a closely related mechanism in the context of coups. Importantly, AHP show that public learning from regime survival can undo the uniqueness properties generated by static global games (see also Shadmehr (2019).) My technical contribution is to move private noise to participation costs; this, combined with support conditions that preserve two-sided dominance after every public history, restores uniqueness in a repeated global game even with public learning over regime strength.

The model is also related to work on how rebel groups choose among different tactics or technologies of contention. In particular, Bueno de Mesquita (2013) takes a very

similar approach to modeling tactics: he treats conventional and irregular tactics as technologies that differ in how effectively they convert mobilization into political power. In his model, organizational capacity is exogenous and tactical choice follows participation; here, mobilization is endogenously pinned down by beliefs and learning. Qiu (2022) likewise studies dynamic tactical choice, and shows that guerrilla warfare allows rebels to survive and accumulate strength, creating incentives to postpone conventional confrontation until conditions improve. Here, failed protests can undermine future mobilization, creating incentives to deploy participation-intensive tactics early. Morris and Shadmehr (2024) study the choice of repertoire *width*, but focus on which set of contentious actions is made available in a one-shot environment rather than on tactical sequencing across repeated attempts.

Another model of dynamic tactical choice in protest is developed by Boyer et al. (2026). Dynamics in their model, however, arise from changes in the ideological composition of the movement: online mobilization initially draws in large numbers of participants, but also facilitates the emergence of radical factions whose usage of violence subsequently crowds out moderates. In my model, participant preferences are fixed, and a single tactic is assigned to all participants. Hence, tactical change arises not from learning about ideological composition, but from learning about regime strength and consequent changes in the relative productivity of tactics.

Finally, the empirical implications of my model connect to substantive research on the determinants and consequences of contentious politics. For instance, organizational resources and structural conditions affect the forms of contention available to movements (Blair 2024; Dube and Vargas 2013; Sonin and Wright 2024). Other research emphasizes how tactical choices shape participation and public support: nonviolent resistance may lower barriers to participation (Chenoweth and Stephan 2011), while violence can discourage subsequent participation or change how outside audiences respond to a movement (Manekin and Mitts 2022; Simpson, Willer and Feinberg 2018; Steinert-Threlkeld, Chan and Joo 2022; Wasow 2020). I abstract from these direct effects on participation costs, movement composition, and outside audiences to isolate a different mechanism: how the prospect of demobilization following setbacks shapes the sequencing of tactics, even when citizens' underlying preferences, the distribution of participation costs, and the available tactical repertoire remain unchanged.

3 Model

Overview. There are two periods, $t \in \{1, 2\}$, and two sets of strategic actors: a protest organizer and a unit mass of potential participants. The organizer chooses the tactic used in each period in order to maximize the probability that the movement succeeds over its lifetime. Whenever the organizer is indifferent between the two tactics, he chooses peace.[1] The organizer is forward-looking and does not discount the future. The organizer also cannot compel participation: after observing the organizer's choice of tactic, citizens make individually optimal participation decisions.

The movement confronts a regime of unknown strength θ. Before the game begins, θ is drawn Unif$[0, M]$, with $M > 1$. Regime strength is fixed across periods. Let F_t denote the commonly held beliefs at the beginning of period t, so $F_1 = \text{Unif}[0, M]$. A successful protest ends the game; following a failed period 1 protest, the movement may try again in period 2.

Tactics and protest power. In each period in which the game remains active, the organizer chooses between two tactics: *peaceful* and *violent*, denoted by $\tau \in \{p, v\}$. Each tactic is associated with a continuously differentiable and weakly increasing production function $A_\tau(\ell)$, which maps turnout $\ell \in [0, 1]$ – the share of potential participants who join the protest – into protest power. A protest using tactic τ succeeds if and only if $A_\tau(\ell) > \theta$. For all $\ell \in [0, 1]$, assume:

(A1) $A_v(0) > A_p(0) = 0$: violence has higher baseline power;

(A2) $A_v(1) < A_p(1) = 1$: peace has higher full-participation power;

(A3) $A_v'(\ell) < A_p'(\ell)$: peaceful power is more responsive to additional participation.

A1 captures the idea that violent tactics can generate some power even with very little mass participation. A2 says that at the opposite end of full mobilization, peaceful tactics are more powerful. A3 ensures that this difference arises systematically as turnout increases. Together, the assumptions imply that the two technologies cross exactly once: there exists a unique $\ell^* \in (0, 1)$ such that $A_v(\ell) > A_p(\ell) \quad \text{for } \ell < \ell^*$, and vice versa for $\ell > \ell^*$.

Participation. Each potential participant i privately observes a period-t participation cost

$$c_{it} = c_t + \epsilon_{it}$$

[1] The tie-breaking assumption is needed only at knife-edge parameter values or histories at which the organizer is exactly indifferent between the two tactics.

where $c_t \sim \text{Unif}[-\epsilon, 1+\epsilon]$ is a common component of participation costs and $\epsilon_{it} \sim \text{Unif}[-\epsilon, \epsilon]$ is idiosyncratic. The common shocks c_t are independent across periods. The idiosyncratic shocks ϵ_{it} are i.i.d. across agents and periods. Citizens do not observe c_t or ϵ_{it} directly. The support of c_t guarantees two-sided dominance and ensures that conditioning on being the marginal participant does not truncate the support of the idiosyncratic shock. In particular, every possible participation cutoff lies at least ϵ away from the boundaries of the common-shock support.

A citizen who participates receives benefit 1 if the protest succeeds and pays their participation cost regardless of the outcome. Thus,

$$u_{it} = \begin{cases} 1 - c_{it}, & \text{if } i \text{ participates and the protest succeeds,} \\ -c_{it}, & \text{if } i \text{ participates and the protest fails,} \\ 0, & \text{if } i \text{ does not participate.} \end{cases}$$

Timing. In each period in which the game remains active:

1. Given the public history and current beliefs F_t, the organizer chooses a tactic $\tau \in \{p, v\}$.
2. Participation costs are realized and privately observed. Citizens simultaneously decide whether to participate.
3. The protest occurs. Turnout ℓ_t and the protest outcome are publicly observed.
4. If $A_\tau(\ell_t) > \theta$, the protest succeeds and the game ends. If the protest fails in period 1, beliefs are updated and the game proceeds to period 2. The game ends after period 2 regardless of the outcome.

The solution concept is Perfect Bayesian Equilibrium.

3.1 Discussion of model assumptions

The model focuses on one channel through which failed protests affect subsequent mobilization: by revealing discouraging information about regime strength, as in AHP. This choice reflects the paper's substantive interest in how organizers anticipate and adapt to endogenous demobilization. In practice, failures may also encourage organizational learning, strengthen activists' commitments, or motivate renewed recruitment efforts, potentially increasing future participation.[2] Such mechanisms lie outside the scope of

[2]For a formal model in which a committed core left behind by initial failure facilitates subsequent mobilization, see De Mesquita and Shadmehr (2023).

this model, which aims instead to provide a microfoundation for demobilization following setbacks and examine its consequences for tactical choice.

Finally, I assume that participation costs are independently drawn from the same distribution in each period. This abstracts from persistently committed "core activists" and from conditions that generate autocorrelation in costs. The assumption ensures that period-2 participation responds to the history of the game through updated beliefs about regime strength rather than through persistence in cost realizations. Allowing such persistence would introduce an additional source of intertemporal dependence, distinct from the learning about regime strength studied here.

The assumption of tactic-independent participation costs also serves to isolate the mechanism of interest. In practice, some forms of participation, particularly those associated with violent tactics, may entail greater physical, legal, or psychological costs. The model deliberately abstracts from these differences, showing that organizers may prefer to begin peacefully even when violence is no more costly and offers a greater probability of immediate success. Moreover, differences in tactical effectiveness can equivalently be represented as tactic-specific participation costs. As I show in as shown in Appendix A.2, for any given belief state, one can hold the tactical production function fixed and introduce tactic-specific cost differences that reproduce exactly the same equilibrium individual participation decisions.

4 Power and the informational cost of failure

In this section, I abstract away from tactical choice to isolate the feedback loop between protest power, beliefs, and future mobilization. By removing the role of the tactic-choosing organizer, I leave the movement fully exposed to the informational consequences of failure. Because it must use the same protest technology in both periods, it cannot adapt the way it converts participation into power after beliefs and mobilization deteriorate.

The core of the model is that more power is valuable because it increases the probability of defeating the regime today, but, conditional on failure, it produces worse news – since the regime survives a strong challenge, citizens learn that it must itself be relatively strong. In the model, since a protest succeeds if and only if $A(\ell_t) > \theta$, following a failure, agents learn with certainty that $\theta \geq A(\ell_t)$, ruling out all weaker regime types. The common posterior belief is thus obtained by truncating the prior distribution from

below at the realized power of the failed protest.[3]

To show how this change in beliefs weakens the movement's future mobilization capacity, I first characterize how it factors into individual participation decisions. Citizens join the protest whenever the expected benefit of success exceeds their private cost. As Lemma 1 shows, the participation decision takes a cutoff form:

Lemma 1 (Cutoff participation)**.** *There exists a threshold $\hat{c}_t$ such that agent i participates in the period-t protest if and only if $c_{it} < \hat{c}_t$, where*

$$\hat{c}_t = \int_0^1 F_t\left(A(\ell)\right) d\ell. \tag{1}$$

Equation (1) gives the first link in the power–learning feedback: Equation (1) gives the first link in the power–learning feedback: public beliefs determine the highest participation cost a citizen is willing to bear and still join the protest. The threshold of the decision is derived from the participation decision of the marginal participant, who averages over the turnout levels that might be generated, and asks how likely the corresponding protest power is to defeat the regime.[4] In that sense, the participation cutoff is simply an average of expected success, given the fixed tactic $A(.)$ and current beliefs F_t. A pessimistic shift in current beliefs therefore *decreases* the participation cutoff, and in turn, shrinks the set of citizens whose costs will fall below it.

The next step is to translate the participation cutoff into realized turnout:

Lemma 2 (Equilibrium turnout)**.** *Given participation cutoff $\hat{c}_t$, realized turnout is*

$$\ell_t = \begin{cases} 1 & \text{if } c_t \leq \hat{c}_t - \epsilon \\ \frac{\hat{c}_t - c_t + \epsilon}{2\epsilon} & \text{if } \hat{c}_t - \epsilon < c_t < \hat{c}_t + \epsilon \\ 0 & \text{if } c_t \geq \hat{c}_t + \epsilon \end{cases} \tag{2}$$

Define

$$\rho := \frac{1}{1 + 2\epsilon} \tag{3}$$

[3]This sharp truncation is stronger than the mechanism requires. The key is the "bad news" *direction* of learning. With noisier or more gradual learning, the feedback from failure to beliefs and future mobilization would operate in the same direction.

[4]The derivation utilizes the standard Laplacian beliefs property of global games (Morris and Shin 2003). Conditional on being marginal, the agent is uncertain only about her rank in the population cost distribution; as the common shock varies uniformly over $(\hat{c}_t - \epsilon, \hat{c}_t + \epsilon)$, her rank – and hence induced turnout – ranges linearly from 0 to 1, yielding a uniform distribution of turnout over $[0, 1]$. The result relies on the uniform distribution of the common shock c_t, but not on uniform idiosyncratic noise – any continuous idiosyncratic distribution yields the same uniform rank belief.

Then $\Pr(\ell_t = 1) = \rho\hat{c}_t$, $\Pr(\ell_t = 0) = \rho(1 - \hat{c}_t)$, *and with remaining probability* $1 - \rho$, *turnout is in* $(0, 1)$ *and is uniformly distributed conditional on being interior. Hence, expected turnout is*

$$\mathbb{E}\left[\ell_t \mid \hat{c}_t\right] = \frac{1-\rho}{2} + \rho\hat{c}_t \tag{4}$$

Although citizens are forward-looking, their participation decisions are *as if myopic*, as in AHP. Because each citizen is atomistic, her current action cannot affect regime survival or the continuation history, so continuation payoffs cancel from the participation decision.

One feature of the turnout process will be significant for the analysis of tactical choice. As Equation 4 shows, ρ governs how strongly changes in beliefs are amplified into aggregate mobilization. When ρ is high, "noise" in citizens' participation propensities around the common shock is very tightly clustered. As such, small decreases in $\hat{c}_t$ (which is pinned down by contemporaneous beliefs, F_t) can cause many citizens to cross the participation boundary together, so a deterioration in beliefs can significantly reduce turnout. When ρ is low, the same belief change produces a more muted aggregate response.

The preceding results characterize behavior at any fixed public belief state, and allow the game to be solved recursively. At every history, public beliefs uniquely determine a participation cutoff (Lemma 1); the cutoff and the common cost shock then uniquely determine turnout and protest power (Lemma 2). Applying these steps yields a unique equilibrium path:

Proposition 1 (Unique cutoff PBE)**.** *There exists a unique Perfect Bayesian Equilibrium in threshold strategies. At every on-path history, agents participate in period* t *if and only if* $c_{it} < \hat{c}_t$ *with* $\hat{c}_t$ *given by* (1)*. Beliefs update by truncation following failure. Given any realization of the common shocks* $\{c_t\}_{t=1}^{2}$*, these equilibrium strategies induce a unique path of turnout, protest outcomes, and posterior beliefs.*

The uniqueness result is useful in its own right and does not rely on the two-period horizon. Because atomistic citizens' continuation values cancel from their current participation decisions, the cutoff argument extends to an arbitrary number of periods or an infinite horizon (provided continuation payoffs are well-defined). As AHP show, uniqueness does not follow simply from repeating a stage game with a unique equilibrium. In their setting, learning from regime survival can eliminate the lower-dominance region after an initial failed attack, restoring equilibrium multiplicity. In my model, the cost-signal structure, together with the support assumption that preserves two-sided

dominance after every public history, pins down a unique participation cutoff in every period. The informational consequences of failure can therefore be analyzed without additional equilibrium selection arguments.

Corollary 1 (Failure erodes mobilization)**.** *Following a failed protest,*

$$\hat{c}_{t+1} \leq \hat{c}_t \qquad \textit{and} \qquad \mathbb{E}\left[\ell_{t+1}\right] \leq \mathbb{E}\left[\ell_t\right]$$

where both inequalities are strict whenever failure produces a nontrivial pessimistic update that affects the success probability of the tactic.

The corollary states the dynamic implication of the learning process on mobilization: expected mobilization falls over time. Note, however, that *expected* decline does not preclude *realized* increases in turnout. Because the common shock is re-drawn each period, a particularly low period-2 draw can produce a turnout increase.

Since the informational cost of failure increases with the power of the failed challenge, should movements sometimes prefer a weaker protest? In the fixed-tactic benchmark, the answer is no. Greater power comes with a greater informational cost upon failure – however, the two forces do not grow at the same rate. Increasing protest power initially comes with a substantial informational cost: if the protest still fails, the additional power rules out many relatively weak regimes that would otherwise have remained plausible. But this cost tapers off as the protest becomes stronger. Once failure has already selected heavily toward strong regimes, there is progressively less favorable continuation value left for an additional increase in power to destroy, while that additional power continues to expand the set of regimes that can be defeated immediately. In other words, the movement's lifetime probability of success is convex in realized protest power.

Proposition 2 (Higher realized power improves lifetime success)**.** *Fix a single tactic A that is used in both periods. The movement's probability of succeeding by the end of period 2, conditional on realized period-1 power x, is weakly increasing and convex in x.*

The first-order implication of the result is that more power is always better – the informational downside is never large enough to make less power desirable. Convexity states that the value of additional power actually rises with the strength of the protest – a second-order implication about the *marginal* value of power. Additional power is least valuable when the movement is weak and becomes increasingly valuable as the protest grows stronger. Equivalently, holding average first-period power fixed, the movement weakly prefers a more dispersed distribution of protest power – one that generates more very weak and very strong realizations rather than concentrating probability on

intermediate ones. The implication is that intermediate levels of power face the worst power-learning trade-off: they offer only limited upside in terms of power, while their failure would create substantial bad news. Hence, an intermediate realization of power is less attractive than an appropriate lottery over weaker and stronger realizations with the same mean. This observation is particularly relevant for the analysis of tactical choice when $\rho \to 1$ (section 5.1.2).

The fixed-tactic benchmark of this section therefore fully exposes the movement to the power-learning trade-off. Since the same technology determines both current protest power and the continuation state, there is no way to hedge against or mitigate the damaging effects of failure. The next section introduces tactical choice as precisely such an instrument.

5 Tactical choice

I now re-introduce the organizer and study how he optimally chooses tactics. The organizer is forward-looking, but cannot commit in advance to a tactical sequence. At each point, he evaluates tactics by their effect on the movement's *lifetime* chance of success. For the first period, this means that tactical choice depends upon each tactic's comparative advantage across *future* mobilization states, as well as upon which tactic generates more power today.

Schematically, let x denote the realized power of the first-period protest. The organizer's problem is

$$\max_{\tau_1 \in \{p,v\}} \mathbb{E}_{x|\tau_1} \Bigg[\underbrace{\Pr(\text{success now} \mid x)}_{\text{immediate success}} + \underbrace{\Pr(\text{failure now} \mid x)}_{\text{reach period 2}} \cdot \underbrace{\max_{\tau_2 \in \{p,v\}} \Pr(\text{success later} \mid \text{failure at } x, \tau_2)}_{\text{optimal continuation}} \Bigg]$$

The expectation is taken over the distribution of first-period power induced by the chosen tactic and equilibrium participation. Following failure, the organizer observes x and chooses the tactic that maximizes the probability of success under the updated beliefs and mobilization conditions.

The organizer's problem resembles a variant of standard dynamic experimentation. In the typical experimentation setting, a decision-maker chooses among actions or tech-

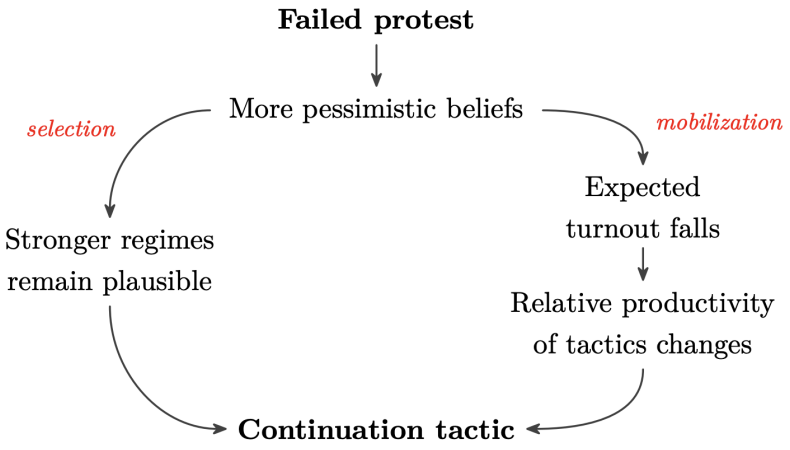


Figure 1. Two channels through which failure shapes tactical choice

nologies not only for their immediate payoff, but also for what their outcomes reveal, and how that information improves subsequent choices. This creates a tradeoff between exploiting the action that currently appears best and experimenting, or exploring, in order to learn. In my setting, however, experimentation comes with an additional feedback loop: since tactics are turnout-dependent technologies, the information generated by a protest changes not only beliefs, but also the future productivity of the available technologies. Experimentation can therefore degrade the environment in which future exploitation occurs: a failed protest may leave the organizer better informed, but facing lower mobilization and hence a different set of tactically valuable options.

To understand this trade-off, I first make two general observations about both the mobilization and selection incentives present in the model. Regarding the mobilization pathway, I observe that a tactic can be worth postponing for future use because it is *not* degraded by learning-from-failure. To see why, suppose an organizer has access to a violent technology which has arbitrarily high baseline power, but no turnout sensitivity at all. The following lemma states that the organizer never uses it first.

Lemma 3 (Flat violence is held in reserve)**.** *If $A_v(\ell) \equiv \bar{v}$ for all ℓ, then a success-maximizing organizer never uses violence in period 1.*

The logic is that of preservation. A completely turnout-insensitive tactic – such as some pre-positioned sabotage that can be triggered without any people on the scene – loses no power from being postponed, as its power is always $\bar{v}$. By contrast, the peaceful tactic is *perishable:* its effectiveness deteriorates with falling turnout. It is thus most valuable at the start of the game, and only weakens (in expectation) after any failure. The organizer therefore uses the perishable tactic first, and preserves the

turnout-insensitive tactic for a potential, more pessimistic future.

As the corollary states, violence can attract more participants, generate more expected power and have a higher probability of immediate success, yet still optimally be held in reserve. Its robustness to demobilization creates *option value* that is lost when it is used immediately.

Corollary 2 (Immediate advantage need not imply early use)**.** *Suppose violence is turnout-insensitive, so that $A_v(\ell) \equiv \alpha$, and $\alpha > 1/2$. Then violence attracts greater expected participation, generates greater expected first-period power, and is more likely to succeed immediately than peace. Nevertheless, the organizer strictly prefers to open peacefully.*

The second observation pertains to the selection channel. Preserving a turnout-insensitive tactic may be valuable from the perspective of the first period, but will it actually be used in the continuation? The answer is not necessarily. Because peace has the higher ceiling of power, sufficiently severe failures can leave behind precisely the strong regimes against which peace has the comparative advantage. A tactic that is valuable to preserve when regime strength is uncertain may cease to be useful once failure has selected sufficiently strongly toward powerful opponents. The implication is an asymmetry in tactical paths: some movements may optimally remain peaceful after every history, but no movement remains violent after every history (a sufficient condition for peace to be used on the entire path is stated in Appendix Proposition A1.) After a sufficiently powerful failure, only the higher-ceiling peaceful technology remains well suited to the strong regime types left in the posterior.

The remainder of the section uses a linear setting to translate these sequencing forces into concrete predictions about how observable features of movements and their tactical repertoires affect the trajectories that movements actually adopt.

5.1 Linear tactics

In the remainder of the paper, I normalize the peaceful technology to $A_p(\ell) = \ell$, and let

$$A_v(\ell) = \alpha + \beta\ell, \qquad \alpha > 0, \quad \beta \geq 0, \quad \alpha + \beta < 1.$$

The normalization is without loss of generality within the linear setting: the slope of peaceful power simply fixes the units in which protest power and regime strength are measured.

This parameterization makes each tactic's comparative advantage especially transparent. The intercept α captures violence's low-turnout advantage: unlike peace, violence can generate power even with little or no participation. By contrast, the slope difference $1-\beta$ captures peace's advantage at scaling with turnout. Equivalently, β measures how closely violence keeps pace with peace as additional participants join. Thus, higher α strengthens violence's low-turnout advantage, whereas higher β weakens peace's turnout-based advantage. The restriction $\alpha + \beta < 1$ preserves peace's higher ceiling under full mobilization.

These parameters also lend themselves to empirical interpretation. Violence's baseline capacity α captures the extent to which small groups can generate leverage with low participation. α is high when actors can exert pressure by disrupting strategically or symbolically important targets, such as infrastructure, centers of state power, or centers of economic activity. Access to capital, rugged terrain, or poorly protected targets can all raise α. Violence's turnout sensitivity β captures the extent to which violent contention itself benefits from numbers. Some forms of violence are inherently manpower-intensive: coordinated assaults, attempts to seize and hold government buildings or territory, sustained barricade defense, and direct-fire engagements become more effective as more participants can be mobilized. Other tactics, such as sabotage, assassination, or some forms of bombing and indirect fire, can be carried out effectively by relatively small groups and therefore depend much less on mass participation.

The linear specification also makes the fundamental objects of the game particularly simple to compute. Under the initial uniform prior,

$$\hat{c}_1^p = \frac{1}{2M}, \qquad \hat{c}_1^v = \frac{\alpha + \beta/2}{M}.$$

Hence, for $\rho > 0$,

$$\mathbb{E}\left[\ell_1^v\right] > \mathbb{E}\left[\ell_1^p\right] \iff \alpha + \frac{\beta}{2} > \frac{1}{2}.$$

Violence can therefore attract *more* participants when its technological advantages make success sufficiently plausible. Peace has no intrinsic participation advantage in the model: relative turnout is itself an equilibrium outcome of tactical effectiveness.[5]

A static benchmark follows immediately. A myopic organizer would therefore simply

[5] Allowing peaceful participation to be intrinsically less costly would mechanically shift participation toward peace and, all else equal, strengthen incentives for peaceful protest. I abstract from this channel to isolate differences in participation generated endogenously by tactical effectiveness and beliefs.

choose the tactic generating greater expected first-period power, which is given by

$$S_1^\tau = \frac{\mathbb{E}\left[A_\tau(\ell_1^\tau)\right]}{M}$$

where ℓ_1^τ denotes equilibrium first-period turnout under tactic τ. Corollary 2 already showed how the dynamic problem can overturn this ranking: a tactic can attract more participants, generate more expected power, and be more likely to succeed immediately, yet still optimally be held in reserve.

5.1.1 Responses to setbacks

I first characterize tactical choice after a failed protest of size x. A natural conjecture is that more severe setbacks should make violence increasingly attractive: Since these failures have the largest demobilizing effect, they should also make violence – the tactic advantaged at low turnout – most attractive. The model delivers the opposite prediction: when violence is used in continuation, it follows sufficiently *low-power* failures. By contrast, high-power failures lead the movement back to peace.

Proposition 3 (Continuation tactics)**.** *There exists a cutoff $c_\rho(\alpha, \beta; M) \in [0, \alpha + \beta)$ such that violence is strictly preferred in period 2 following a failed first-period protest if and only if*

$$x < c_\rho(\alpha, \beta; M).$$

Whenever $c_\rho > 0$, the organizer is indifferent at $x = c_\rho$. The cutoff is weakly increasing in both α and β.

The result is driven by *selection*: larger failures progressively eliminate the weaker regime types against which violence has a comparative advantage. Let ℓ^* denote the turnout level at which the two technologies cross. In the linear specification, $\ell^* = \frac{\alpha}{1-\beta}$, and because $A_p(\ell^*) = A_v(\ell^*) = \ell^*$, this is also the power level associated with the technological crossing. The continuation cutoff satisfies $c_\rho(\alpha, \beta; M) < \ell^*$. Thus, the organizer switches back to peace before realized failure power reaches the power level associated with the technological crossing.

The reason is that tactical choice depends on the entire posterior over regime strength, not just its weakest remaining type. Following a failure of size $x < \ell^*$, violence retains an advantage against the relatively weak regimes with $\theta \in [x, \ell^*)$. But the posterior *also* contains all stronger regimes with $\theta \geq \ell^*$, against which peace requires less turnout to succeed. As x rises toward ℓ^*, the set of regime types for which violence is useful shrinks, while the set for which peace has the advantage remains. Peace therefore becomes optimal strictly before x itself reaches the technological crossing point.

This also bears implications for participation. Once $x \geq \ell^*$, the participation cutoff under peace is weakly higher than under violence (that is, $\hat{c}_2^p(x) \geq \hat{c}_2^v(x)$.) The intuition is simple. At low turnout levels, where violence would normally have a technological advantage, neither tactic now generates enough power to defeat any regime remaining in the posterior. Thus, violence's advantage is irrelevant for citizens' participation decisions. At the higher turnout levels where success remains possible, peace is the stronger technology, and draws more expected participants.

The comparative statics of the cutoff identify which movements are most likely to use violence after a setback. Higher baseline capacity α expands the range of failures after which violence is useful. Higher β does the same for a different reason: as violence becomes more responsive to turnout, peace loses some of its scaling advantage – thus pushing the cutoff point upwards and also expanding the range of histories for which violence can be optimal in the continuation.

Finally, the continuation choice of tactics is not always path dependent. When violent capacity is sufficiently weak – that is, it is not attractive even after the weakest possible failure – the cutoff collapses to $c_\rho(\alpha, \beta; M) = 0$, so peace is optimal after every possible failure. As violent capacity improves, the cutoff moves into the interior, and continuation choice becomes history-dependent.

5.1.2 Choosing an opening tactic

The continuation problem conditions on the power x generated by an initial failed protest. The opening problem is different: before turnout is realized, the organizer's choice of tactic helps determine the *distribution* of first-period power, and hence the set of belief and mobilization states from which the continuation game may begin. The parameter ρ is central to this comparison because it determines how strongly the information revealed by the first protest feeds back into subsequent mobilization. I begin with the two limiting cases of ρ to isolate the mobilization channel clearly.

Resilient mobilization: $\rho \to 0$. When ρ approaches zero, changes in public confidence have vanishingly small effects on realized turnout. Although failure still changes beliefs about regime strength and citizens' willingness to participate, these changes no longer appreciably shift the distribution of aggregate mobilization. Turnout approaches $\text{Unif}[0, 1]$ at every belief state. This limit serves to study tactical choice in the absence of the mobilization feedback loop.

	Opening tactic	Following failure
Weak violent capacity	Peace	Peace
Medium violent capacity	Peace	Violence after weak failure; peace after strong failure
Strong violent capacity	Violence	Violence after weak failure; peace after strong failure

Table 1. Equilibrium tactics by violent capacity when $\rho \to 0$

With the mobilization feedback muted, the organizer can place greater weight on which tactic performs best under favorable current mobilization. Sufficiently scalable violence can therefore be used immediately. The model admits three equilibrium trajectories, summarized in Table 1.[6] To understand the forces that generate these trajectories, I plot the two relevant boundaries in (α, β) space (Figure 2). The orange boundary governs continuation tactical choice, and represents the lowest level of β at which $c_0(\alpha, \beta; M) > 0$, so violence becomes an appealing fallback tactic. The higher green boundary $\beta^*(\alpha)$ governs the opening decision, and is the $\rho \to 0$ case of the general opening threshold $\beta^*_\rho(\alpha; M)$ characterized in Proposition 4 below.

The figure also clarifies how the two dimensions of violent capacity affect the equilibrium trajectory. When violent capacity is low on either dimension, never used on-path: violence is simply a worse version of the peaceful tactic. Increasing baseline capacity α alone cannot make violence an appealing opening tactic. Note that when $\beta = 0$, increasing α only makes violence a useful reserve tactic, but does not make violence a useful opening tactic (reproducing Lemma 3). When $\alpha > 0$, increasing β first makes violence attractive after weak failures. Further increases in β improve violence disproportionately in high-turnout states; after crossing $\beta^*(\alpha)$, this is strong enough to justify using violence immediately. Thus, α and β can both shift the movement toward violence, but through different parts of the sequence: α primarily strengthens its fallback role, while β strengthens its appeal under favorable mobilization. The two dimensions are in a sense complementary: α makes violence worth retaining as a fallback, while β can make that capacity valuable enough under high mobilization to deploy immediately.

[6] There is no corresponding region in which the organizer opens with violence but switches to peace after every possible failure. If violence would not remain optimal following even the weakest failed violent opening, then it cannot be strictly optimal to deploy first. Details are given in Appendix B.2.

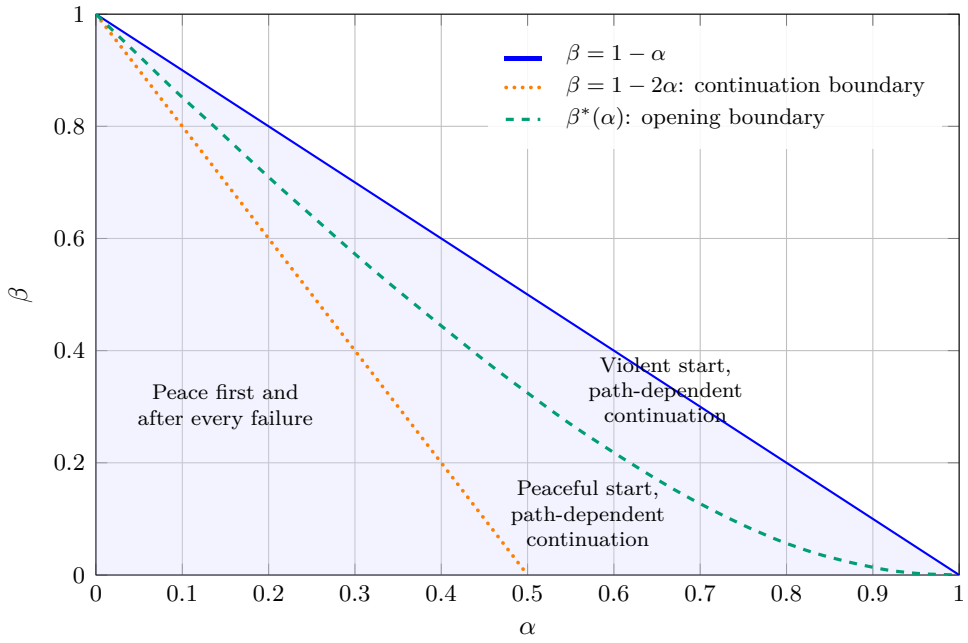


Figure 2. Tactical choice when $\rho \to 0$. The solid blue line marks the boundary of the feasible parameter region, $\beta < 1-\alpha$, which is shaded blue. The orange dotted line, $\beta = 1-2\alpha$, is the continuation boundary: below it, peace is optimal after every possible failure; above it, violence is optimal following sufficiently weak failures. The green dashed line, $\beta^*(\alpha)$, is the opening boundary. It is computed numerically by fixing a grid of values of α and, for each value solving for the value of β at which the organizer is indifferent between opening peacefully and opening violently, i.e., $W_1^p(\alpha, \beta; M) = W_1^v(\alpha, \beta; M)$.

Fragile mobilization: $\rho \to 1$. This case makes the mobilization channel as strong as possible. Since potential participants respond very similarly to even tiny changes in beliefs, realized turnout becomes concentrated near either zero or full mobilization – the interior component of the turnout distribution vanishes. Because of this turnout property, realized peaceful protest power approaches an all-or-nothing distribution: the movement generates either essentially no power or the maximum power it can produce.

This produces a threshold-like aggregate technology even though the peaceful production function itself remains linear. Classic threshold accounts of collective action emphasize how clustered participation thresholds can produce abrupt changes in aggregate mobilization (Granovetter 1978; Kuran 1991). A related pattern arises here through the responsiveness of participation to common beliefs: when participation propensities are sufficiently concentrated, small changes in confidence can move the movement sharply between near-zero and near-full mobilization. Thus, observed peaceful power may look

highly discontinuous even though the underlying technology is continuous.

This polarization echoes the convexity result in Proposition 2. In the fixed-tactic benchmark, holding average first-period power fixed, greater dispersion improves lifetime success. When $\rho \to 1$, peace generates precisely such polarized outcomes: approximately 0 or 1. It thus offers an unusually attractive opening option. By using peace first, the movement gains all the power upside of a strong challenge (if everyone turns out) and none of the informational downside of a failed weak challenge (if nobody turns out).

Violence, on the other hand, is doubly disadvantaged as an opening tactic, since it compresses realized power toward the interior, approximately α or $\alpha+\beta$. Its baseline capacity $\alpha > 0$ means that even a poorly mobilized violent failure is guaranteed to produce pessimistic information about the regime, while the restriction $\alpha + \beta < 1$ prevents even full violent mobilization from matching peace's upper bound. When changes in beliefs feed strongly into subsequent turnout, this difference in the distribution of first-period power has especially large dynamic consequences.

Note the contrast with the $\rho \to 0$ benchmark. A high β still creates a technological reason to use violence immediately: favorable early mobilization conditions are precisely when violence is most powerful. But when $\rho \to 1$, the preservation motive dominates even at the highest feasible β. Violence can still be optimal following sufficiently weak failures, but it is never used as an opening tactic. The three trajectories from the resilient-mobilization benchmark therefore collapse to two.

Intermediate levels of mobilization. The following proposition generalizes the limiting-case comparison. Above a critical threshold of ρ, violence is never used as an opening tactic, no matter how effective.

Proposition 4 (Mobilization responsiveness and violent openings)**.** *For every (α, M), there exists a unique threshold $\bar{\rho}(\alpha, M) \in (0, 1)$ such that:*

(i) If $\rho < \bar{\rho}(\alpha, M)$, sufficiently scalable violence is used first. In particular, there exists a unique threshold $\beta^*_\rho(\alpha; M) < 1 - \alpha$ such that the organizer opens violently if and only if

$$\beta > \beta^*_\rho(\alpha; M).$$

(ii) If $\rho \geq \bar{\rho}(\alpha, M)$, the organizer opens peacefully for every admissible violent technology.

Equilibrium trajectory	$\rho \to 0$	$\rho \to 1$	Dependence on ρ
Peace throughout	Possible	Possible	Region shrinks with ρ
Peace first; violence after weak failures, peace after strong failures	Possible	Possible	Region expands in between extreme values of ρ
Violence first; violence after weak failures, peace after strong failures	Possible	Absent	Region disappears above a critical ρ

Table 2. Equilibrium tactical trajectories and mobilization responsiveness.

Thus, holding the tactical repertoire fixed, changes in the responsiveness of mass participation can generate discrete changes in the trajectories available to a movement, and in particular in whether violence can appear at the outset.

Table 2 summarizes how mobilization responsiveness reshapes equilibrium tactical trajectories. First, as ρ increases, the region in which movements remain exclusively peaceful shrinks. Higher ρ makes turnout more polarized, increasing the second-period appeal of tactics that retain leverage even with very few participants. Second, the region in which movements begin peacefully but turn to violence after weak failures expands between the two limiting cases. This expansion occurs from both directions. The first direction is the shrinking of the exclusively peaceful region as violence becomes attractive as a fallback. The other direction is the shrinking of the violence-first region disappears: even highly effective violent tactics are worth preserving for later rather than deploying immediately. The result is a wider range of circumstances in which organizers begin peacefully while reserving violence for sufficiently weak failures.

5.1.3 Structural change and tactical trajectories

The preceding results offer a theory of how broader structural changes reshape the trajectories of contention. As scholars of comparative politics have long observed, urbanization, changes in infrastructure and communications, transformations of public space, and shifts in the social composition of movements all shape movements' repertoires of contention – and in particular, how effectively different tactics convert turnout into political leverage. The model implies that such changes can affect not only which tactics are viable, but *when* they are used.

Urbanization provides the clearest example. The movement of revolutionary con-

tention toward cities concentrated populations, economic activity, and centers of political authority in the same space, while also bringing challengers into closer contact with the coercive apparatus of the state (Beissinger 2022; Wallace 2014). One plausible consequence is to *rotate* the violent technology (Kalyvas and Balcells 2010): urbanization reduces the leverage that a small armed group can exercise independently of mass support (Fearon and Laitin 2003), while increasing the extent to which coercive tactics benefit from large numbers (Chenoweth and Stephan 2011; Staniland 2010). In the language of the linear setting, α falls while β rises.

The effect on trajectories is a rise in early-stage violence. A decline in α erodes violence's comparative advantage after demobilization: weakly attended movements can no longer generate as much leverage simply by switching tactics. An increase in β makes violent confrontation increasingly effective when turnout is already high. Urbanization can therefore shift violence *forward* in the tactical sequence even while reducing its value as a fallback.

The reconstruction of Paris under Haussmann is a localized example of the same logic. The narrow streets of the old city made barricades comparatively effective instruments for small groups of insurgents (Traugott 2010). After the Haussmannian renovation, broad boulevards and large, open spaces facilitated troop movement and rendered barricade insurrection more difficult and less effective (Hobsbawm 2005; Traugott 2010). From the perspective of the model, this meant that successful street confrontation could no longer be easily sustained by a small group of insurgents defending barricades, and required assembling and sustaining larger numbers. Thus, to the extent that the transformation both lowers α and raises the relative importance of β, it should reduce the incidence of violence as a post-demobilization fallback and concentrate whatever violent contention remains in periods when mobilization is already broad.

The development of critical infrastructure can move the tactical environment in the opposite direction. Rail lines, energy systems, communications networks, ports, and other concentrated nodes can create opportunities for a relatively small number of actors to impose costs far beyond their numbers (Mitchell 2011; Silver 2003; Wright 2000). Where such systems are vulnerable to disruption (Tichý and Eichler 2018), technological development can raise α without comparably increasing β. The model predicts that such changes should move violence *backward* in the sequence. Higher baseline capacity expands the set of post-failure states in which violence is useful, making peaceful-to-violent transitions more likely after weak setbacks. But precisely because this capacity survives demobilization, it need not be spent while participation is high. The prolifer-

ation of vulnerable strategic targets can therefore increase a movement's capacity for violence while making that capacity disproportionately visible *after* mass mobilization falters. Decentralizing or protecting those same nodes should have the opposite effect: it removes violence's low-turnout insurance value and shrinks the region of peaceful-to-violent switching.

Structural change can also reshape the movement's mobilization base itself: how much do supporters differ in their response to changing prospects of success? Durable parties, unions, neighborhood organizations, and activist networks can produce a layered constituency, where not all citizens respond identically to disappointments. Such heterogeneity dampens the aggregate response to any common deterioration in confidence, corresponding to a lower ρ. The opposite environment can produce much sharper swings in mobilization. This possibility is especially familiar from work on protest under authoritarian rule, where repression and preference falsification can leave citizens uncertain about both the extent of latent opposition and the regime's vulnerability (Kuran 1991; Lohmann 1994). When participation depends heavily on these common assessments, a single political signal can move large numbers of otherwise similar citizens across the participation margin at once. What may appear empirically as a volatile "mass psychology" can therefore arise from highly synchronized responses to common information, rather than from unusually fickle individual preferences. In the language of the model, such "high ρ" movements can surge rapidly when success appears plausible, but can also collapse rapidly following discouraging failures.

6 Repression

I have so far treated the movement's tactical environment as fixed. In practice, states actively intervene in that environment. In this section, I study how repression interacts with two forces at the center of the baseline model: tactical substitution and learning from failure. While characterizing the regime's fully unrestricted dynamic repression policy is outside the scope of this study, I analyze two benchmark policy problems, each designed to isolate one of these channels. The first studies how the movement's ability to substitute across tactics constrains the effects of responsive repression – that is, measures taken after an initial protest has already occurred. The second holds responsive repression fixed in order to isolate how preemptive repression – enacted before any protest has happened – changes the impact of failures on future mobilization.

Formally, I introduce a regime actor whose objective opposes that of the organizer: the regime receives payoff 1 if it survives the movement and 0 if the movement succeeds.

In each period, the regime can pay some cost to weaken tactic τ by a factor $\gamma_t^\tau \in [0,1]$, so that realized protest power is $x_t = \gamma_t^\tau A_\tau(\ell_t)$, (so lower γ denotes more intense repression.) I make several assumptions: First, repression is publicly observed before participation decisions, so citizens account for it both when deciding whether to participate and when interpreting a subsequent failure. Second, I focus on *tactic-specific* interventions which affect one tactic, and leave the alternative technology unchanged. Finally, I assume that the regime shares the movement's uncertainty about its underlying strength.[7]

I begin by analyzing the limits to responsive repression enacted in the terminal period. At this point, the regime only targets the tactic which the protest will use. As repression progressively weakens this tactic, however, the alternative eventually becomes more attractive. Once the organizer switches, further repression of the abandoned tactic has no additional effect on tactical choice or the movement's probability of success – effectively placing a cap on the value of repression.

Proposition 5 (Repression and tactical substitution)**.** *Suppose that, following some first-period failure, the organizer strictly prefers one tactic while the alternative tactic remains viable. If the regime represses only the preferred tactic, there is a unique repression threshold at which the organizer switches to the alternative. Once that threshold is reached, further repression of the original tactic cannot further reduce the movement's probability of success.*

Notably, this demonstrates one reason why tactical adjustment patterns may differ from those described in the previous section. Absent repression, weak failures lead toward violence while stronger failures lead toward peace. Repressing violence following a weak failure can reverse this logic by inducing a switch toward peace. Repressing peace following a stronger failure can analogously redirect the movement toward violence whenever violence remains a viable fallback. Thus repression can generate *opposite patterns* of tactical adjustment from pure learning.

I now turn to *preemptive* repression enacted before the first protest. Such intervention weakens the power of the movement's immediate challenge, but *also* changes what that failure reveals about regime strength. A weaker failure can therefore leave the movement more optimistic, and better able to mobilize, in the period that follows. To isolate this informational consequence, I hold period-2 repression fixed at zero. The previous subsection already characterizes how the regime can respond once a failure has occurred; this benchmark isolates how preemptive repression changes the state from which that

[7]Unobserved repression or private regime information would introduce additional inference and signaling dynamics that I abstract from here.

continuation begins. Because weaker failures leave movements more optimistic, repression consequently returns part of its immediate benefit to the movement through more favorable continuation beliefs:

Proposition 6 (The belief refund)**.** *Suppose period-2 technologies are unrepressed. Lowering realized first-period protest power weakly improves the movement's continuation prospects. Moreover, for sufficiently weak first-period challenges ($x \leq \min\{\alpha, c_\rho(\alpha, \beta; M)\}$), lowering first-period power leaves the movement's eventual probability of success unchanged when $\rho\beta = 0$, but strictly* increases *eventual success when $\rho > 0$ and $\beta > 0$.*

The proposition is conditional on realized first-period power. It characterizes the informational consequences of making a protest weaker, rather than the organizer's ex ante response to a particular repression policy, which also changes the distribution of first-period outcomes.

The belief refund can do more than erase the regime's immediate gain from repression: when mobilization responds to beliefs, it can make repression *counterproductive*. The contrast is clearest in the two limiting cases. When $\rho = 0$, turnout is insensitive to beliefs, and repression can receive a *full refund* in the region of sufficiently weak first-period challenges. In this region, failure induces a switch to violence, whose baseline capacity is sufficient to defeat the marginal regime types spared by repression in period 1. When $\rho > 0$, however, repression can protect and even bolster mobilization. By weakening the first-period challenge, the regime makes a subsequent failure less discouraging. Because turnout responds to beliefs, the resulting optimism translates into higher second-period participation. Crucially, that additional participation can make the movement's fallback tactic more powerful. When $\beta > 0$, violence becomes more effective as more citizens participate. If turnout is high, the violent tactic can defeat the marginal regime types that repression spared in period 1, and even threaten still stronger regimes. The refund is then *more than complete*: reducing the movement's immediate power can increase its eventual probability of success.

This is the regime-side counterpart of the preservation motive in the previous section. There, a forward-looking organizer values robust violence because failure destroys the mobilization conditions under which peaceful protest is effective. Preemptive repression partly prevents that destruction: by weakening a failure, the regime actually saves the optimism and mobilization that the movement would otherwise have lost.

The same informational logic has a direct implication for the persistence of violence. Consider a regime that represses an anticipated violent opening. As long as the organizer

continues to open violently, stronger counterinsurgency makes the initial protest both more likely to fail, and to fail *weakly*. Weak failures generate precisely the states in which violence remains optimal in the continuation.

Proposition 7 (Weakening versus displacing violence)**.** *Suppose repression affects only the period-1 violent technology and there is no period-2 repression. Among repression policies under which the organizer continues to open violently, stronger repression weakly increases the expected number of periods of violent protest. By contrast, any policy strong enough to induce a peaceful opening produces strictly less expected violence than any policy that preserves a violent opening.*

Repressing violence can therefore have different effects depending on whether it actually prompts the movement to change course. If repression is not too strong and the movement continues to begin violently, stronger repression shifts failures toward precisely the continuation states in which violence recurs. Once repression becomes strong enough to displace the violent opening, the certain first-period episode of violence disappears.

7 Discussion

This paper develops a model of dynamic tactical choice in repeated protests. Its broader contribution is to identify a form of political experimentation in which experimentation is informative about the environment but also changes the productivity of the actions available to the decision-maker. Here, protest outcomes reveal information about regime strength, but also feed back into subsequent mobilization, which in turn determines the effectiveness of the tactics available to the movement. Experimentation can therefore erode the value of some technologies while preserving others. This creates incentives to hold robust tactics in reserve, to switch tactics following some failures but not others, and even to forgo the tactic with the greatest immediate prospect of success. The relevant dynamic tradeoff is thus not simply between current success and future information, but between current success and the political environment in which future contention will take place.

Repression introduces a second strategic actor into this problem. State intervention does more than reduce the immediate effectiveness of a protest: it changes the information generated by failure and can redirect the movement toward tactics that remain effective under the resulting mobilization conditions. Repression is therefore an intervention in a dynamic process of learning and tactical substitution, rather than simply a reduction in challengers' current power. This helps explain why the regime's ability to weaken a particular protest need not translate into greater long-run security, and why

flexibility in responding to future contention can itself alter the movement's incentives in the present.

A fuller analysis of the regime's intertemporal problem presents one avenue for future work. In particular, greater policy flexibility need not benefit the regime, because movements choose tactics in anticipation of future state responses. If an organizer expects violence to be heavily repressed once it becomes an attractive fallback after failure, the value of preserving violence for that contingency falls, potentially inducing the movement to deploy it earlier. A regime that could credibly commit to future forbearance might therefore induce a peaceful opening by relinquishing an intervention it would prefer to use ex post. This raises a broader commitment problem: restrictions on the regime's future actions may improve its position today by changing the movement's incentives over how to sequence tactics.

The same informational logic also suggests a role for the public transmission of protest outcomes. If failed protests demobilize partly through what they reveal about regime strength, then obscuring the scale of a strong but unsuccessful challenge can blunt the discouraging information generated by its failure. Conversely, a regime may sometimes benefit from allowing opposition setbacks to become widely observed, even when greater transparency also reveals the strength of the challenge itself. A natural extension would therefore allow the regime to shape the precision or dissemination of information about protest outcomes, connecting the model to work on information control and the informational foundations of repression.

The institutional organization of tactical choice provides another extension. The unitary organizer studied here offers a useful benchmark, but contested leadership and collective decision-making may substantially alter the incentives to experiment with, preserve, or switch between tactics. With multiple leaders, tactical sequences may depend on bargaining power, veto rights, opportunities for unilateral action, and the possibility of organizational fragmentation. Under more decentralized forms of decision-making, tactical choice may instead depend on changes in the composition of the movement itself, as different constituencies enter, exit, or gain influence following success and failure. Extending the analysis to these alternative decision structures would help clarify how organizational institutions mediate the dynamic incentives identified here.

More generally, the model points toward a class of experimentation problems in which actions are technologies whose productivity depends on an endogenous state that past outcomes themselves reshape. Protest tactics provide one instance, but similar

forces may arise whenever political actors choose among instruments that differ in their dependence on participation, organizational capacity, public support, or other resources that political experience can subsequently erode or replenish. A broader theory of such problems could characterize these incentives in a more general experimental environment, extending some of the insights developed in the more specific case studied here. The protest setting therefore present a special case of a broader dynamic problem in which actors choose amongst technologies whose future value depends on the conditions that today's actions help create.

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Appendix

Table of Contents

A Proofs for participation and benchmark results **A1**

A.1 Participation and turnout A1

A.2 Participation-equivalent cost representation A4

A.3 Realized power and lifetime success A5

A.4 Benchmark results with tactical choice A6

B Linear setting **A8**

B.1 Continuation choice (Proposition 3) A8

B.2 The opening comparison A12

B.3 Proof of Proposition 4 . . . . . A15

B.4 Algebraic verifications A16

C Proofs for repression extension **A21**

C.1 Responsive repression A21

C.2 Preemptive repression and the belief refund A23

C.3 Persistence of violence A25

A Proofs for participation and benchmark results

A.1 Participation and turnout

Proof of Lemma 1. Fix beliefs F_t and a tactic with production function A. Because each citizen is atomistic, her current participation decision has no effect on whether the regime survives, the posterior inherited by the next period, or the continuation game. Continuation payoffs therefore cancel from the comparison between participating and abstaining, so participation is as-if myopic.

Suppose other citizens use cutoff $\hat{c}_t$. A higher realization of c_{it} shifts citizen i's posterior over the common cost c_t upward, which weakly lowers expected turnout and hence the expected benefit from participating. Since the cost of participation also rises one-for-one in c_{it}, the payoff difference between participating and abstaining is strictly decreasing in c_{it}. It is therefore enough to characterize the marginal citizen who is indifferent at $c_{it} = \hat{c}_t$.

Conditional on $c_{it} = \hat{c}_t$, $c_t = \hat{c}_t - \epsilon_{it}$. Since $\hat{c}_t \in [0, 1]$ and $c_t \sim \text{Unif}[-\varepsilon, 1 + \varepsilon]$, the interval $[\hat{c}_t - \varepsilon, \hat{c}_t + \varepsilon]$ lies inside the support of the common shock. Conditioning on marginality therefore does not truncate the support of ϵ_{it}. Moreover, because both c_t and ϵ_{it} have constant densities on their supports, the marginal citizen's posterior over c_t is uniform on $[\hat{c}_t - \varepsilon, \hat{c}_t + \varepsilon]$.

For any c_t in this interval, aggregate turnout under cutoff $\hat{c}_t$ is

$$\ell(c_t; \hat{c}_t) = \Pr(c_t + \epsilon_{jt} < \hat{c}_t) = \frac{\hat{c}_t - c_t + \varepsilon}{2\varepsilon}.$$

Thus, as the marginal citizen's posterior over c_t ranges uniformly over $[\hat{c}_t - \varepsilon, \hat{c}_t + \varepsilon]$, induced turnout ranges uniformly over $[0, 1]$. Her expected benefit from participating is therefore

$$\int_0^1 F_t(A(\ell))\, d\ell.$$

Indifference of the marginal citizen gives

$$\hat{c}_t = \int_0^1 F_t(A(\ell))\, d\ell,$$

which is equation (1). Since the right-hand side lies in $[0, 1]$, it defines a valid cutoff; by the monotonicity above, citizens with $c_{it} < \hat{c}_t$ participate and those with $c_{it} > \hat{c}_t$ abstain. □

Proof of Lemma 2. Fix a participation cutoff $\hat{c}_t \in [0,1]$. Actual turnout is

$$\ell_t = \begin{cases} 1, & c_t \le \hat{c}_t - \varepsilon, \\ \dfrac{\hat{c}_t - c_t + \varepsilon}{2\varepsilon}, & \hat{c}_t - \varepsilon < c_t < \hat{c}_t + \varepsilon, \\ 0, & c_t \ge \hat{c}_t + \varepsilon. \end{cases}$$

The support of c_t has length $1 + 2\varepsilon$. The set of common shocks generating full turnout has length $\hat{c}_t$, so

$$\Pr(\ell_t = 1) = \frac{\hat{c}_t}{1 + 2\varepsilon} = \rho \hat{c}_t.$$

The set generating zero turnout has length $1 - \hat{c}_t$, so

$$\Pr(\ell_t = 0) = \rho(1 - \hat{c}_t).$$

The remaining interval has length 2ε and hence probability $1 - \rho$. Conditional on this interior region, the mapping from c_t to ℓ_t is linear, so realized turnout is uniformly distributed on $(0,1)$. Consequently,

$$\mathbb{E}[\ell_t \mid \hat{c}_t] = \rho \hat{c}_t + (1 - \rho)\frac{1}{2} = \frac{1 - \rho}{2} + \rho \hat{c}_t.$$

□

Lemma 2 implies that for a fixed tactic A, its probability of success under beliefs F_t is defined as follows:

Definition A1 (Equilibrium success probability)**.** *For beliefs F_t and tactic A, denote by $S(A, F_t)$ the actual probability of success. This is given by:*

$$S(A, F_t) = \rho(1 - \hat{c}_t) F_t(A(0)) + (1 - \rho) \int_0^1 F_t(A(\ell))\, d\ell + \rho \hat{c}_t F_t(A(1)).$$

Since

$$\hat{c}_t = \int_0^1 F_t(A(\ell))\, d\ell,$$

this can equivalently be written as

$$S(A, F_t) = \rho(1 - \hat{c}_t) F_t(A(0)) + (1 - \rho)\hat{c}_t + \rho \hat{c}_t F_t(A(1)).$$

Proof of Proposition 1. At every history, beliefs F_t and the tactic in use uniquely determine citizens' participation cutoff through equation (1). Given this cutoff and a realization of the common shock c_t, Lemma 2 uniquely determines realized turnout and therefore realized protest power. Following a failed protest, Bayes' rule uniquely determines the truncated posterior.

Backward induction therefore uniquely determines citizens' threshold strategies and the organizer's optimal tactic at every history, with indifference resolved by the tie-breaking convention in favor of peace. Hence, for every realization of the underlying state and common shocks, equilibrium strategies induce a unique path of turnout, protest outcomes, and posterior beliefs. □

Proof of Corollary 1. Suppose a period-t protest fails at realized power x. The posterior F_{t+1} is obtained by truncating F_t from below at x. Hence

$$F_{t+1}(y) \leq F_t(y) \qquad \text{for every } y,$$

with strict inequality on a nonempty interval whenever the failure produces a nontrivial update. Equation (1) then implies

$$\hat{c}_{t+1} = \int_0^1 F_{t+1}(A(\ell))\, d\ell \leq \int_0^1 F_t(A(\ell))\, d\ell = \hat{c}_t.$$

The inequality is strict whenever the posterior shift affects a positive-measure set of power realizations generated by the tactic.

By Lemma 2, expected turnout is

$$\mathbb{E}[\ell_t \mid \hat{c}_t] = \frac{1-\rho}{2} + \rho \hat{c}_t.$$

which is strictly increasing in $\hat{c}_t$. The result therefore follows immediately from the statement that $\hat{c}_{t+1} \leq \hat{c}_t$. □

A.2 Participation-equivalent cost representation

Lemma A1 (Participation-equivalent cost representation)**.** *Fix a belief state F_t and tactical technologies A_p and A_v. Let $\hat{c}_t^p$ and $\hat{c}_t^v$ denote the equilibrium participation cutoffs under peace and violence, respectively.*

Define

$$k_p(F_t) = 0, \qquad k_v(F_t) = \hat{c}_t^p - \hat{c}_t^v.$$

Consider an alternative specification in which both tactics use the same protest technology A_p, but participation under tactic τ entails cost $c_{it} + k_\tau(F_t)$. This specification generates exactly the same equilibrium participation cutoffs, and hence the same participation decisions and turnout distributions, as the original model, conditional on F_t and the chosen tactic.

Proof. In the original model, Lemma 1 establishes that

$$\hat{c}_t^\tau = \int_0^1 F_t(A_\tau(\ell))\, d\ell.$$

In the alternative specification, let $\tilde{c}_t^\tau$ denote the equilibrium cutoff under tactic τ. Since both tactics now use A_p, the marginal citizen's indifference condition is

$$\tilde{c}_t^\tau + k_\tau(F_t) = \int_0^1 F_t(A_p(\ell))\, d\ell = \hat{c}_t^p.$$

Therefore,

$$\tilde{c}_t^p = \hat{c}_t^p,$$

and

$$\tilde{c}_t^v = \hat{c}_t^p - k_v(F_t) = \hat{c}_t^p - (\hat{c}_t^p - \hat{c}_t^v) = \hat{c}_t^v.$$

Thus the equilibrium participation cutoffs are identical underthe two specifications. □

The implied cost differential has a simple interpretation: if violence offers a lower expected probability of success than peace, then $\hat{c}_t^v < \hat{c}_t^p$, so $k_v(F_t) > 0$: from the perspective of individual participation decisions, violence behaves as though it were more costly. Conversely, if violence offers a higher expected probability of success, then $k_v(F_t) < 0$, and violence behaves as though it were less costly. Thus, differences in tactical effectiveness can be interpreted as implicit participation-cost differences, although the sign and magnitude of these differences depend on prevailing beliefs.

As a caveat, the cost-differential specification needs not generate the same protest outcomes or subsequent beliefs: tactic-specific technologies directly affect realized protest power, whereas participation costs operate through citizens' decisions to participate. The equivalence therefore does not extend to the organizer's dynamic optimization problem.

A.3 Realized power and lifetime success

Proof of Proposition 2. Fix a single increasing tactic A that is used in both periods, and let $\bar{A} := A(1)$. Let $V(x)$ denote the movement's probability of succeeding by the end of period 2 conditional on realized period-1 power x.

Recall that after a failure at power x, the posterior is Unif$[x, M]$. Define

$$h(x) := \int_0^1 (A(\ell) - x)_+ \, d\ell, \qquad G(x) := \int_0^1 \mathbf{1}\{A(\ell) > x\} \, d\ell.$$

Intuitively, $h(x)$ measures the tactic's average power above x, while $G(x)$ is the share of turnout realizations for which the tactic generates power above x. Note also that function h is absolutely continuous, with $h'(x) = -G(x)$ almost everywhere.

Recall that the period-2 participation cutoff is given by

$$\hat{c}_2(x) = \frac{h(x)}{M - x}.$$

Full turnout succeeds with probability

$$r(x) := \frac{\bar{A} - x}{M - x}.$$

Using Lemma 2, the actual period-2 success probability is therefore

$$S_2(x) = (1 - \rho)\hat{c}_2(x) + \rho \hat{c}_2(x) r(x).$$

Under the prior Unif$[0, M]$, lifetime success conditional on period-1 power x is

$$\begin{aligned} V(x) &= \frac{x}{M} + \frac{M - x}{M} S_2(x) \\ &= \frac{x}{M} + \frac{M - x}{M} \hat{c}_2(x) \Big[1 - \rho + \rho r(x)\Big] \\ &= \frac{x}{M} + \frac{h(x)}{M} \Big[1 - \rho + \rho r(x)\Big] \\ \implies \qquad MV(x) &= x + h(x) \Big[1 - \rho + \rho r(x)\Big] \end{aligned} \tag{A1}$$

Since

$$r'(x) = -\frac{1 - r(x)}{M - x},$$

differentiating (A1) and substituting that $h'(x) = -G(x)$:

$$MV'(x) = 1 - G(x) + \rho\Big(1 - r(x)\Big)\Big(G(x) - \hat{c}_2\Big). \tag{A2}$$

All terms on the right-hand side are weakly positive. Hence $V'(x) \geq 0$ almost everywhere, implying that V is weakly increasing.

We now turn to the convexity claim. Let

$$s(x) := 1 - r(x) = \frac{M - \bar{A}}{M - x}.$$

Then

$$s'(x) = \frac{s(x)}{M - x}, \qquad \hat{c}_2'(x) = -\frac{G(x) - \hat{c}_2(x)}{M - x} \quad \text{a.e.}$$

As x rises, fewer turnout realizations satisfy $A(\ell) > x$, so $G(x)$ is weakly decreasing. Because G may have jumps, it is not necessarily legal to generally differentiate it pointwise everywhere. Instead, we use its distributional derivative: the decline in G is represented by the nonnegative measure $-dG$. Taking the distributional derivative of (A2) gives

$$M\, dV'(x) = \Big[1 - \rho s(x)\Big](-dG) + \frac{2\rho s(x)}{M - x}\Big(G(x) - \hat{c}_2(x)\Big)\, dx. \tag{A3}$$

Both terms on the right-hand side are nonnegative. $-dG$ is nonnegative because G is weakly decreasing; moreover, reachable power satisfies $x \leq \bar{A}$, so $s(x) \in [0, 1]$ and $1 - \rho s(x) \geq 0$, while $G(x) - \hat{c}_2(x) \geq 0$ as shown above. Therefore the distributional second derivative of V is a nonnegative measure, which implies that V is weakly convex over the range of reachable first-period power. □

A.4 Benchmark results with tactical choice

Proposition A1 (A sufficient condition for peace on the entire path)**.** *Suppose peace is optimal following every possible period-1 failure. Let X_τ denote realized period-1 protest power when the organizer chooses tactic $\tau \in \{p, v\}$, under the equilibrium turnout distribution induced by that tactic.*

If furthermore for every z,

$$\mathbb{E}\left[(X_p - z)_+\right] \geq \mathbb{E}\left[(X_v - z)_+\right], \tag{A1.1}$$

then peace is optimal in period 1 as well.

Proof of Proposition A1. Let $V_p(x)$ denote the movement's probability of succeeding by the end of period 2 conditional on realized period-1 power x and peaceful continuation following failure. By Proposition 2, V_p is weakly increasing and convex. Condition (A1.1) requires the peaceful opening to generate at least as much expected power above every threshold z as the violent opening. For bounded random variables, this condition implies

that every increasing convex function takes weakly higher expected value under X_p than under X_v.Hence, applying it to the increasing convex function V_p,

$$\mathbb{E}[V_p(X_p)] \geq \mathbb{E}[V_p(X_v)].$$

Because peace is optimal after every possible failure, these are exactly the lifetime success probabilities from opening peacefully and violently. Thus peace is optimal in period 1. □

Proof of Lemma 3. Suppose $A_v(\ell) \equiv \bar{v}$. By the assumptions on the technologies, $\bar{v} \in (0, 1)$. If the organizer opens with violence and the protest fails, the posterior places probability 1 on $\theta \geq \bar{v}$. Flat violence can then never succeed, so the optimal continuation following a violent opening is peace.

Compare this $v \to p$ sequence with the feasible strategy $p \to v$. For every $\theta < \bar{v}$, both strategies eventually succeed: under $p \to v$, either peace succeeds immediately or the reserved violent tactic does so in period 2. For $\theta \geq \bar{v}$, violence cannot succeed under either sequence, so success depends entirely on the peaceful attempt. Under $p \to v$, that attempt occurs under the initial beliefs; under $v \to p$, it occurs only after failure at power $\bar{v}$, which strictly lowers the peaceful participation cutoff. By Lemma 2, using the same realization of the common cost shock therefore produces weakly lower turnout, and hence weakly lower peaceful power, when peace is postponed, with strict inequality with positive probability.

Since $v \to p$ is optimal conditional on opening with violence, whereas after opening peacefully the organizer can do weakly better than the restricted $p \to v$ strategy, it follows that the organizer strictly prefers using peace in period 1 rather than flat violence. □

B Linear setting

This appendix establishes the results for the linear specification $A_p(\ell) = \ell$ and $A_v(\ell) = \alpha + \beta\ell$, where $\alpha > 0$, $\beta \geq 0$, and $\alpha + \beta < 1$. The argument has three main parts:

First, I characterize the organizer's continuation choice after a failed first-period protest and identify the two continuation boundaries β_0 and β_α. Second, I establish the shape of the period-1 opening comparison: violence cannot be strictly optimal before it becomes repeatable, while after it becomes repeatable its opening advantage is strictly increasing in β. Third, these results reduce the proof of Proposition 4 to the opening comparison as β approaches the feasibility boundary $1 - \alpha$.

Some of the sign arguments needed for the first two parts are algebraically involved, but contribute little to the overall logic or intuition of the proof. I collect those calculations in Section B.4. Throughout the main argument, $\rho \in (0, 1)$, as implied by the model. Values $\rho = 0$ and $\rho = 1$ are used below only as continuous limiting cases.

B.1 Continuation choice (Proposition 3)

Following a failed first-period protest of realized power x, the posterior is Unif$[x, M]$. Let $S_p(x)$ and $S_v(x)$ denote the actual period-2 probabilities of success under peace and violence, respectively. It is useful to derive these objects explicitly, because the violent continuation problem changes form depending on whether the posterior lower bound x lies below or above violence's floor α.

Let F_x denote the posterior CDF and let $\hat{c}_2^\tau(x)$ denote the period-2 participation cutoff under tactic τ. From Lemma 1,

$$\hat{c}_2^\tau(x) = \int_0^1 F_x(A_\tau(\ell))\, d\ell.$$

Given a cutoff $\hat{c}_2^\tau(x)$, Lemma 2 implies that the actual success probability can be written as

$$S_\tau(x) = \rho(1 - \hat{c}_2^\tau(x))F_x(A_\tau(0)) + (1 - \rho)\hat{c}_2^\tau(x) + \rho\hat{c}_2^\tau(x)F_x(A_\tau(1)).$$

The three terms correspond, respectively, to zero turnout, an interior turnout realization, and full turnout. The middle term simplifies because, conditional on interior turnout, ℓ is uniform on $(0, 1)$ and the average success probability is exactly the participation cutoff $\hat{c}_2^\tau(x)$.

For peace, $A_p(\ell) = \ell$. Since a failed first-period protest has $x \geq 0$,

$$\hat{c}_2^p(x) = \int_x^1 \frac{\ell - x}{M - x}\, d\ell = \frac{(1 - x)^2}{2(M - x)}, \qquad F_x(A_p(0)) = 0, \qquad F_x(A_p(1)) = \frac{1 - x}{M - x}.$$

substituting yields

$$S_p(x) = \frac{(1-x)^2}{2(M-x)}\left[(1-\rho) + \rho\frac{1-x}{M-x}\right]. \tag{A4}$$

For violence, there are three relevant cases.

Case 1: Suppose $x \leq \alpha$. Even zero turnout then produces power at least as large as the posterior lower bound, so

$$\hat{c}_2^v(x) = \int_0^1 \frac{\alpha + \beta\ell - x}{M-x}\, d\ell = \frac{\alpha + \beta/2 - x}{M-x}.$$

Using

$$F_x(A_v(0)) = \frac{\alpha - x}{M-x}, \qquad F_x(A_v(1)) = \frac{\alpha + \beta - x}{M-x},$$

and simplifying yields

$$S_v(x) = \frac{\alpha - x + (1-\rho)\beta/2}{M-x} + \frac{\rho\beta(\alpha + \beta/2 - x)}{(M-x)^2}, \qquad x \leq \alpha. \tag{A5}$$

Case 2: Suppose $\alpha < x < \alpha + \beta$. Then, low-turnout violent protests generate power below the posterior lower bound and therefore have no chance of success. Only turnout above $(x-\alpha)/\beta$ matters, so

$$\hat{c}_2^v(x) = \int_{(x-\alpha)/\beta}^1 \frac{\alpha + \beta\ell - x}{M-x}\, d\ell = \frac{(\alpha + \beta - x)^2}{2\beta(M-x)}.$$

Here $F_x(A_v(0)) = 0$ and $F_x(A_v(1)) = (\alpha + \beta - x)/(M-x)$, which gives

$$S_v(x) = \frac{(\alpha + \beta - x)^2}{2\beta(M-x)}\left[(1-\rho) + \rho\frac{\alpha + \beta - x}{M-x}\right], \qquad \alpha < x < \alpha + \beta. \tag{A6}$$

Case 3: Finally, if $x \geq \alpha + \beta$, even full-turnout violence cannot defeat any regime type remaining in the posterior, so $S_v(x) = 0$.

Proof of Proposition 3. I show that the continuation advantage of violence is single-crossing in the power x of the failed first-period protest: as x rises, the organizer can switch at most once from violence to peace.

First, note that if $x > \alpha + \beta$, violence cannot succeed at any level of mobilization. For the remaining cases, I proceed in two steps, which are summarized below.

Case 1. In the intermediate region $x \in [\alpha, \alpha + \beta)$. I will show that $S_v(x)/S_p(x)$ – the relative advantage of violence over peace – is decreasing in x

Case 2. If $x \leq \alpha$, then the baseline power of violence is sufficient to defeat some regime types in the posterior. In this region, I will show that the baseline capacity α required for indifference is increasing in x.

The two regions then combine to give a single cutoff from violence to peace.

Case 1: Let $\beta > 0$ and consider $x \in [\alpha, \alpha + \beta)$.

Dividing (A6) by (A4) gives

$$\frac{S_v(x)}{S_p(x)} = \frac{1}{\beta}\left(\frac{\alpha+\beta-x}{1-x}\right)^2 \frac{(1-\rho)(M-x)+\rho(\alpha+\beta-x)}{(1-\rho)(M-x)+\rho(1-x)}. \tag{A7}$$

Both nonconstant ratios on the right-hand side are strictly decreasing in x. For the first, this follows from $\alpha + \beta < 1$. For the second, write it as $(C - x)/(D - x)$, where $C = (1 - \rho)M + \rho(\alpha + \beta)$ and $D = (1 - \rho)M + \rho$, so $C < D$. Hence $S_v(x)/S_p(x)$ is strictly decreasing on this region.

Case 2: Consider $x \leq \alpha$.

In this case, the ratio S_v/S_p is less useful because violence's baseline capacity lies *inside* the posterior support. Instead, fix (x, β, M, ρ) and consider how large α must be to make the organizer indifferent. Let $\alpha^*(x)$ denote this indifference value. Section B.4.1 shows that $\alpha^*(x)$ is strictly increasing in x, while $S_v(x) - S_p(x)$ is strictly increasing in α. Thus, for any fixed actual α, violence is preferred exactly when $\alpha > \alpha^*(x)$, and this inequality can switch from true to false at most once as x rises.

The two comparisons fit together continuously at $x = \alpha$. In particular, both expressions for violent success give

$$S_v(\alpha) = \frac{\beta}{2(M-\alpha)}\left[(1-\rho) + \rho\frac{\beta}{M-\alpha}\right].$$

Moreover, at $x = \alpha + \beta < 1$, violence has zero probability of success, while peace still has positive probability of success, i..e $S_v(\alpha+\beta) = 0 < S_p(\alpha+\beta)$. Thus the comparison over the two regions combines into a single global cutoff $c_\rho(\alpha, \beta; M) \in [0, \alpha + \beta)$, such that violence is strictly preferred if and only if $x < c_\rho$.

Note that when $\beta = 0$, the region $(\alpha, \alpha + \beta)$ is empty. The same low-x argument applies for $x \leq \alpha$, while $S_v(x) = 0$ for $x \geq \alpha$, yielding the same cutoff characterization.

It remains to establish the comparative statics of the cutoff. Fix a posterior state x. Raising either α or β makes the violent technology pointwise weakly stronger. Its

participation cutoff therefore weakly rises. Coupling the two environments using the same realization of the common participation-cost shock, the stronger technology induces weakly higher turnout and, because the technology itself is also stronger, weakly higher realized protest power. Thus $S_v(x)$ weakly increases with both α and β, while $S_p(x)$ is unchanged. The set of posterior states at which violence is optimal can therefore only expand, so $c_\rho(\alpha, \beta; M)$ is weakly increasing in both parameters. □

The two continuation boundaries. Having established the organizer's decision rule in the continuation, I proceed to the opening problem. Two cutoff values of β will be useful. The first determines whether violence is ever useful after a failure. By Proposition 3, it is enough to compare the tactics following the weakest possible failure, $x = 0$. From (A4)–(A5), the sign of $S_v(0) - S_p(0)$ is the sign of

$$B_0(\beta) := \rho\beta^2 + [M(1-\rho) + 2\rho\alpha]\,\beta + 2M\alpha - [M(1-\rho) + \rho]\,. \tag{A8}$$

Thus $B_0(\beta) > 0$ exactly when violence is preferred following the weakest possible failure. Since $B_0'(\beta) = 2\rho\beta + M(1-\rho) + 2\rho\alpha > 0$, this comparison moves strictly in violence's favor as β rises.

Define

$$\alpha_0(M, \rho) := \frac{1-\rho}{2} + \frac{\rho}{2M}.$$

At $\beta = 0$, $B_0(0) = 2M\Big(\alpha - \alpha_0(M, \rho)\Big)$. Hence, if $\alpha < \alpha_0(M, \rho)$, there is a unique $\beta_0(\alpha; M, \rho) \in (0, 1-\alpha)$ satisfying $B_0(\beta_0) = 0$. Existence follows from

$$B_0(1-\alpha) = \alpha\Big(M(1+\rho) - \rho\alpha\Big) > 0.$$

For $\beta < \beta_0$, peace is strictly preferred following every possible failure. At $\beta = \beta_0$, the organizer is indifferent only following $x = 0$, while for $\beta > \beta_0$, violence is optimal following sufficiently weak failures. If $\alpha = \alpha_0(M, \rho)$, the boundary is $\beta_0 = 0$; if $\alpha > \alpha_0(M, \rho)$, violence is already useful after sufficiently weak failures at $\beta = 0$, so there is no region in which the continuation tactci is always peace.

The second boundary determines whether violence is *repeatable*, that is, if violence is used in the first period protest (and failed), whether it can ever be followed again by violence.

The weakest violent protest generates power α, so violence is repeatable if $c_\rho > \alpha$. Let $\beta_\alpha(\alpha; M, \rho)$ denote the unique value satisfying $c_\rho = \alpha$. Substituting $x = \alpha$ into the continuation comparison gives

$$\beta\,[(1-\rho)(M-\alpha) + \rho\beta] = (1-\alpha)^2\,[(1-\rho)(M-\alpha) + \rho(1-\alpha)]\,. \tag{A9}$$

The left-hand side is strictly increasing in β and equals zero at $\beta = 0$. At $\beta = 1 - \alpha$ it strictly exceeds the right-hand side. Thus there is a unique $\beta_\alpha(\alpha; M, \rho) \in (0, 1 - \alpha)$.

Finally, note that whenever β_0 exists, $\beta_0 < \beta_\alpha$. At $\beta = \beta_\alpha$, violence and peace are equally attractive following some $x = \alpha > 0$. Single-crossing then implies that violence is strictly preferred following $x = 0$, so it must already have become useful as a fallback at a smaller value of β. Thus β_0 marks the first point at which violence is useful after any failure, while β_α marks the stronger condition under which violence can remain useful after one of its own failed openings.

B.2 The opening comparison

I now characterize period-1 tactical choice. Under the prior, first-period participation cutoffs are

$$\hat{c}_1^p = \frac{1}{2M}, \qquad \hat{c}_1^v = \frac{\alpha + \beta/2}{M}.$$

Let X_p and X_v denote realized period-1 protest power under peaceful and violent openings. Their CDFs on $[0, 1]$ are

$$F_p(x) = \rho(1 - \hat{c}_1^p) + (1 - \rho)x, \qquad 0 \le x < 1, \tag{A10}$$

and

$$F_v(x) = \begin{cases} 0, & x < \alpha, \\ \rho(1 - \hat{c}_1^v) + (1 - \rho)\dfrac{x - \alpha}{\beta}, & \alpha \le x < \alpha + \beta, \\ 1, & x \ge \alpha + \beta. \end{cases} \tag{A11}$$

For some level of realized first-period power x, define

$$H_\tau(x) := \frac{x}{M} + \frac{M - x}{M} S_\tau(x), \qquad \tau \in \{p, v\}.$$

Thus $H_\tau(x)$ is the movement's lifetime success probability conditional on first-period power x and continuation with tactic τ after failure.

Denote the value of the problem to the organizer by

$$V_\beta(x) := \max\{H_p(x), H_v(x)\}.$$

By Proposition 3,

$$V_\beta(x) = \begin{cases} H_v(x) & \text{if } x < c_\rho, \\ H_p(x) & \text{if } x \ge c_\rho, \end{cases}$$

Let $W_1^\tau := \mathbb{E}[V_\beta(X_\tau)]$. This describes (expected) lifetime success conditional upon opening with tactic τ. Finally, define violence's opening advantage by

$$\Delta_\rho(\alpha, \beta; M) := W_1^v - W_1^p.$$

The next lemma states the two facts about Δ_ρ needed for the opening result. If violence is not repeatable, then it also cannot be strictly optimal to use it first. If violence is repeatable, greater turnout sensitivity moves the opening comparison strictly toward violence.

Lemma A2 (Shape of the opening comparison)**.** *Let* $c := c_\rho(\alpha, \beta; M)$.

1. *If* $c \leq \alpha$, *then* $\Delta_\rho(\alpha, \beta; M) \leq 0$. *The inequality is strict whenever* $0 < c \leq \alpha$.
2. *If* $c > \alpha$, *then*
$$\frac{\partial}{\partial \beta} \Delta_\rho(\alpha, \beta; M) > 0.$$

Proof. Proof of part 1. If $\beta = 0$, the result follows from Lemma 3, so suppose $\beta > 0$. If $c \leq \alpha$, every violent opening produces $X_v \geq \alpha \geq c$ and is therefore followed by peace after failure. Hence $W_1^v = \mathbb{E}[H_p(X_v)]$. A peaceful opening, by contrast, is followed by violence after failures with $X_p < c$ and by peace otherwise.

First suppose $0 < c \leq \alpha$. From (A5),

$$H_v(x) = \frac{\alpha + (1-\rho)\beta/2}{M} + \frac{\rho\beta(\alpha + \beta/2 - x)}{M(M-x)} \tag{A12}$$

and therefore

$$H_v'(x) = \frac{\rho\beta(2\alpha + \beta - 2M)}{2M(M-x)^2} < 0. \tag{A13}$$

To compare the two opening choices, separate histories according to whether first-period power falls below or above c. Below c, only a peaceful opening can generate such a weak failure, and the organizer responds by switching to violence. Above c, both opening choices lead to peaceful continuation. Since $H_v(c) = H_p(c)$, there is no discontinuity in value at the boundary. Integrating the two regions by parts gives

$$W_1^p - W_1^v = -\int_0^c F_p(x) H_v'(x)\, dx + \int_c^1 H_p'(x) \left[F_v(x) - F_p(x)\right] dx. \tag{A14}$$

The two terms correspond to these two regions. The first captures the value of the low-power histories that can only be generated by a peaceful opening; it is strictly positive because H_v is decreasing on $[0, c]$. The second compares the distributions of first-period power over histories for which both openings lead to the same peaceful continuation.

The first term is strictly positive by (A13). For the second term, (A10)–(A11) imply that $F_v - F_p$ crosses zero at most once, and only from negative to positive. Moreover, Proposition 2 implies that H_p' is nonnegative and increasing. Section B.4.2 verifies that period-2 indifference at c implies

$$\int_c^1 \left[F_v(x) - F_p(x)\right] dx > 0. \tag{A15}$$

Because the negative part of $F_v - F_p$ occurs below its positive part, weighting by the increasing function H_p' can only strengthen this comparison. Hence the second term in (A14) is nonnegative, and $W_1^p > W_1^v$.

Now suppose $c = 0$. Peace is then weakly optimal following every possible failure, so the relevant conditional lifetime-success function is H_p. Under the initial uniform prior, $S_\tau(0) = \mathbb{E}[X_\tau]/M$, and therefore

$$\int_0^1 [F_v(x) - F_p(x)]\, dx = \mathbb{E}[X_p] - \mathbb{E}[X_v] = M\,[S_p(0) - S_v(0)] \geq 0.$$

Because $F_v - F_p$ crosses at most once from negative to positive, this implies $\int_z^1 [F_v(x) - F_p(x)]\, dx \geq 0$ for every z, or equivalently

$$\mathbb{E}[(X_p - z)_+] \geq \mathbb{E}[(X_v - z)_+] \qquad \text{for every } z.$$

Proposition A1 therefore gives $W_1^p \geq W_1^v$, completing part (i).

Proof of part 2. Suppose $c > \alpha$ and let $\sigma := (c-\alpha)/\beta \in (0,1)$. Thus σ is the interior turnout level under violence that generates exactly the continuation cutoff: $c = \alpha + \beta\sigma$. Violent openings with turnout below σ are followed by violence after failure; those with turnout above σ are followed by peace.

Using V_β, the two opening payoffs can be written as

$$W_1^p = \rho(1 - \hat{c}_1^p)V_\beta(0) + (1-\rho)\int_0^1 V_\beta(x)\, dx + \rho\hat{c}_1^p V_\beta(1), \tag{A16}$$

$$W_1^v = \rho(1 - \hat{c}_1^v)V_\beta(\alpha) + (1-\rho)\int_0^1 V_\beta(\alpha + \beta\ell)\, d\ell + \rho\hat{c}_1^v V_\beta(\alpha+\beta). \tag{A17}$$

When these expressions are differentiated with respect to β, changes in the continuation cutoff generate boundary terms. Those terms cancel because $H_v(c) = H_p(c)$: moving an infinitesimal set of histories across a point at which the organizer is indifferent has no first-order effect on the value.

The remaining calculation is given in Section B.4.3. It yields

$$\frac{\partial \Delta_\rho}{\partial \beta} = (1-\rho)^2 Q_0 + \rho(1-\rho)Q_1 + \rho^2 Q_2, \tag{A18}$$

where Q_0, Q_1, and Q_2 are all strictly positive whenever $c > \alpha$. Since the three weights in (A18) are nonnegative, the derivative is strictly positive. This proves part (ii). □

B.3 Proof of Proposition 4

Lemma A2 reduces the opening problem to a boundary comparison. At $\beta = \beta_\alpha$, we have $c_\rho = \alpha > 0$, so part (i) of the lemma implies

$$\Delta_\rho(\alpha, \beta_\alpha; M) < 0.$$

For every $\beta > \beta_\alpha$, violence is repeatable and part (ii) implies that Δ_ρ is strictly increasing in β. Thus there can be at most one value of β at which the organizer switches from a peaceful to a violent opening. It remains only to determine whether the opening advantage of violence becomes positive before reaching the feasibility boundary $\beta = 1-\alpha$.

Define the boundary comparison

$$\Phi(\alpha, M, \rho) := \lim_{\beta \nearrow 1-\alpha} \Delta_\rho(\alpha, \beta; M).$$

At $\beta = 1 - \alpha$,

$$A_v(\ell) = \alpha + (1-\alpha)\ell \quad \geq \quad \ell = A_p(\ell),$$

with strict inequality whenever $\ell < 1$. Thus violence is point-wise at least as strong as peace. For any posterior following a failure, this also raises the citizens' participation cutoff, so under the same realization of the common participation-cost shock violence induces weakly higher turnout. Because $A_v(\ell) \geq A_p(\ell)$ at every turnout level, realized protest power is therefore weakly higher under violence as well. Hence violence is optimal in period 2 following every failure with power below 1.

At the boundary, violence is used after every failure. For a fixed continuation tactic, lifetime-success values are affine in ρ, while the period-1 turnout distribution is also affine in ρ. Hence $\Phi(\alpha, M, \rho)$ is a polynomial of degree at most two in ρ. Direct calculation yields

$$\Phi(\alpha, M, 0) = \frac{\alpha(1-\alpha)}{6M} > 0, \tag{A19}$$

while

$$\Phi(\alpha, M, 1) = -\frac{\alpha(1-\alpha)(M-1)(2M-\alpha-1)}{4M^3(M-\alpha)} < 0. \tag{A20}$$

It follows that Φ has a root in $(0, 1)$. Furthermore, because (A19)–(A20) have opposite signs, the root is unique.
Denote the root by

$$\bar{\rho}(\alpha, M) \in (0, 1).$$

Hence $\Phi(\alpha, M, \rho) > 0$ for $\rho < \bar{\rho}(\alpha, M)$ and $\Phi(\alpha, M, \rho) < 0$ for $\rho > \bar{\rho}(\alpha, M)$.

Suppose first that $\rho < \bar{\rho}(\alpha, M)$. The opening advantage is strictly negative at β_α and strictly positive at the feasibility boundary. Continuity and strict monotonicity therefore imply a unique

$$\beta^*_\rho(\alpha; M) \in (\beta_\alpha(\alpha; M, \rho), 1 - \alpha)$$

such that $\Delta_\rho(\alpha, \beta^*_\rho; M) = 0$. Thus

$$W_1^v > W_1^p \quad \Longleftrightarrow \quad \beta > \beta^*_\rho(\alpha; M).$$

For $\beta \leq \beta_\alpha$, part (i) of Lemma A2 already rules out a strictly optimal violent opening. This establishes part (i) of Proposition 4.

Now suppose $\rho \geq \bar{\rho}(\alpha, M)$. If $\rho > \bar{\rho}$, the boundary value Φ is strictly negative; if $\rho = \bar{\rho}$, it equals zero. In either case, strict monotonicity on $(\beta_\alpha, 1-\alpha)$ implies

$$\Delta_\rho(\alpha, \beta; M) < 0 \qquad \forall \beta \in (\beta_\alpha, 1-\alpha).$$

Part (i) of Lemma A2 covers the remaining values $\beta \leq \beta_\alpha$. Peace is therefore optimal in period 1 for every admissible violent technology, establishing part (ii).

Additional implications. The same argument gives the ordering of the thresholds. Whenever a violence-first region exists,

$$\beta_\alpha < \beta^*_\rho < 1 - \alpha.$$

Whenever β_0 exists in the feasible parameter space, Section B.1 additionally gives

$$\beta_0 < \beta_\alpha < \beta^*_\rho < 1 - \alpha.$$

The violence-first threshold also approaches the feasibility boundary as mobilization responsiveness approaches its critical value:

$$\lim_{\rho \uparrow \bar{\rho}(\alpha, M)} \beta^*_\rho(\alpha; M) = 1 - \alpha. \tag{A21}$$

To see this, fix any $\varepsilon \in (0, 1-\alpha)$. At $\rho = \bar{\rho}(\alpha, M)$, peace is strictly preferred at every interior $\beta < 1 - \alpha$, so

$$\Delta_{\bar{\rho}}(\alpha, 1 - \alpha - \varepsilon; M) < 0.$$

By continuity in ρ, the same inequality holds for all ρ sufficiently close to $\bar{\rho}$ from below. Since $\Delta_\rho(\alpha, \beta^*_\rho; M) = 0$, it follows that

$$\beta^*_\rho(\alpha; M) > 1 - \alpha - \varepsilon$$

for all such ρ. Because $\beta^*_\rho(\alpha; M) < 1 - \alpha$ always, and ε is arbitrary, (A21) follows.

B.4 Algebraic verifications

This section collects calculations used above used solely to establish a sign.

B.4.1 The low-x continuation comparison

I verify here the claim used in the proof of Proposition 3: for $x \leq \alpha$, the baseline violent capacity required for indifference is increasing in the power of the failed protest.

For fixed (x, β, M, ρ), equations (A4)–(A5) imply

$$\frac{\partial}{\partial \alpha}\left[S_v(x) - S_p(x)\right] = \frac{M + \rho\beta - x}{(M-x)^2} > 0.$$

There is therefore a unique value $\alpha^*(x)$ satisfying $S_v(x) = S_p(x)$, where $\alpha^*(x)$ need not itself lie in the admissible parameter region. Violence is preferred exactly when $\alpha > \alpha^*(x)$. It is therefore enough to show that $\alpha^*(x)$ is increasing in x.

Let

$$p := 1 - x, \qquad k := M - 1, \qquad u := \alpha^*(x) - x.$$

The indifference condition is equivalent to

$$2(k + p + \rho\beta)u = (1-\rho)(k+p)(p^2 - \beta) + \rho(p^3 - \beta^2). \tag{A22}$$

Since $p = 1 - x$ and $u = \alpha^*(x) - x$, $dp/dx = -1$ and $du/dx = d\alpha^*/dx - 1$. Differentiating (A22) with respect to x, and then using the original indifference condition to eliminate u, gives

$$\frac{d\alpha^*(x)}{dx} = \frac{P(p, \beta, k, \rho)}{2(k + p + \rho\beta)^2}, \tag{A23}$$

where

$$\begin{aligned} P(p, \beta, k, \rho) = {} & \rho^2\beta(\beta + 2kp) \\ & + \rho\left[\beta\left\{2k(2-p) + p(4-3p)\right\} + kp(2k+p)\right] \\ & + 2(k+p)^2(1-p). \end{aligned} \tag{A24}$$

Here $k > 0$, $p > 0$, $\beta \geq 0$, and $\rho > 0$. Every displayed term is nonnegative, and the middle line is strictly positive. Hence $d\alpha^*(x)/dx > 0$, as required.

B.4.2 Upper-tail power before repeatability

I verify here equation (A15), the claim needed to sign the second term in (A14). Suppose $0 < c \leq \alpha$ and $S_v(c) = S_p(c)$. Direct integration of (A10)–(A11), followed by use of period-2 indifference at c, gives

$$\int_c^1 \left[F_v(x) - F_p(x)\right] dx = \frac{c}{2M}\left[1 + \rho + c^2 - \rho c - 2\alpha - (1+\rho)\beta\right]. \tag{A25}$$

It remains to show that the bracket is positive.

Let

$$z := 1 - c, \qquad k := M - 1, \qquad u := \alpha - c, \qquad s := z - \beta.$$

Because $c \leq \alpha$ and $\alpha + \beta < 1$, we have $u \geq 0$ and $s - u = 1 - \alpha - \beta > 0$. In these variables, the indifference condition is

$$\begin{aligned} 2(k + z + \rho\beta)u = k(1-\rho)\left[s - z(1-z)\right] - z^2(1-z) \\ + zs(1+\rho) - \rho s^2. \end{aligned} \tag{A26}$$

Using (A26), the bracket in (A25) can be rewritten as

$$\begin{aligned} &1 + \rho + c^2 - \rho c - 2\alpha - (1+\rho)\beta \\ &\quad = \frac{\rho}{k + z + \rho\beta}\Big\{2(k + z + \rho\beta)u + k\left[(1+\rho)s - \rho z(1-z)\right] + zs(1-z)\Big\}. \end{aligned} \tag{A27}$$

Thus it is enough to show $(1+\rho)s - \rho z(1-z) > 0$.

If $s \geq z(1-z)$, this follows immediately. If $s < z(1-z)$, the first term on the right-hand side of (A26) is nonpositive. Since the left-hand side is nonnegative,

$$-z^2(1-z) + zs(1+\rho) - \rho s^2 \geq 0.$$

Hence

$$z(1+\rho)s \geq z^2(1-z) + \rho s^2 > z^2(1-z),$$

where the strict inequality follows from $\rho > 0$ and $s > 0$. Dividing by $z > 0$ gives $(1+\rho)s > z(1-z) \geq \rho z(1-z)$. Thus the desired inequality holds in either case.

All three terms in braces in (A27) are therefore nonnegative, with the final two strictly positive. The bracket in (A25) is strictly positive, proving (A15).

B.4.3 Monotonicity of the opening advantage

I verify here the coefficient signs used in part (ii) of Lemma A2. Throughout this subsection, $c > \alpha$ and $\sigma = (c - \alpha)/\beta \in (0, 1)$.

For the calculation, it is useful to record the conditional lifetime-success functions explicitly. Peaceful continuation gives

$$H_p(x) = \frac{x}{M} + \frac{1-\rho}{2M}(1-x)^2 + \frac{\rho}{2M}\frac{(1-x)^3}{M-x}. \tag{A28}$$

Violent continuation is

$$H_v^L(x) = \frac{\alpha + (1-\rho)\beta/2}{M} + \frac{\rho\beta(\alpha + \beta/2 - x)}{M(M-x)}, \qquad x \leq \alpha, \tag{A29}$$

and

$$H_v^H(x) = \frac{x}{M} + \frac{1-\rho}{2\beta M}(\alpha + \beta - x)^2 + \frac{\rho}{2\beta M}\frac{(\alpha+\beta-x)^3}{M-x}, \qquad \alpha < x < \alpha + \beta. \tag{A30}$$

Differentiating the opening payoffs (A16)–(A17), splitting H_v at $x = \alpha$ and the optimal continuation value at $x = c$, gives boundary terms from the movement of c and σ. These cancel because $H_v(c) = H_p(c)$. After the cancellation and collection of terms in ρ, the derivative takes the form stated in (A18),

$$\frac{\partial \Delta_\rho}{\partial \beta} = (1-\rho)^2 Q_0 + \rho(1-\rho) Q_1 + \rho^2 Q_2.$$

The remainder of the calculation verifies that each coefficient is strictly positive.

Coefficient on $(1-\rho)^2$. Direct integration gives

$$Q_0 = \frac{2\beta + 3(1-\beta)\sigma - 3\alpha\sigma^2 + (1-\beta)\sigma^3}{6M}. \tag{A31}$$

Since $\alpha < 1 - \beta$,

$$\begin{aligned} 6MQ_0 &> 2\beta + (1-\beta)\left(3\sigma - 3\sigma^2 + \sigma^3\right) \\ &= 2\beta + (1-\beta)\left[1 - (1-\sigma)^3\right] > 0. \end{aligned}$$

Coefficient on $\rho(1-\rho)$. Let

$$e := 1 - \alpha - \beta > 0, \qquad k := M - 1 > 0, \qquad u := 1 - \sigma \in (0,1).$$

The derivative of Q_1 with respect to σ simplifies to

$$Q_1'(\sigma) = \frac{kA(u) + B(u)}{2M(k + e + \beta u)^2}, \tag{A32}$$

where

$$\begin{aligned} A(u) &= 2\beta(1-\beta)u^3 + 6\beta e\, u(1-u) + 3e^2(1-u), \\ B(u) &= \beta^2(1-\beta)(u^4 + u^3) + \beta e u^2\left[3\beta + u(2 - 5\beta)\right] \\ &\quad + 6\beta e^2 u(1-u) + 2e^3(1-u). \end{aligned}$$

All terms are nonnegative. The only factor whose sign is not immediate, $3\beta + u(2 - 5\beta)$, is linear in u and positive at both endpoints: $3\beta > 0$ and $3\beta + (2 - 5\beta) = 2(1-\beta) > 0$. Hence $Q_1'(\sigma) > 0$.

Evaluating the coefficient at $\sigma = 0$ and simplifying gives

$$Q_1(\sigma) > \frac{6M\alpha + 4M\beta + 3\alpha^2 + 12\alpha\beta - 12\alpha + 9\beta^2 - 6\beta + 6}{12M^2}. \tag{A33}$$

The numerator is increasing in M. Setting $M = 1$ and using $e = 1 - \alpha - \beta$ reduces it to

$$3 + 10\beta + 3e^2 - 6\beta e = 3 + 3e^2 + \beta(10 - 6e) > 0.$$

Thus $Q_1 > 0$.

Coefficient on ρ^2. Let

$$d := 1 - \alpha - \beta > 0, \qquad D := M - \alpha - \beta > d.$$

Direct simplification gives

$$M^2 Q_2 = T(\beta, d, D),$$

where

$$T(\beta, d, D) = \frac{d^3}{4D} + \frac{\beta(D + \beta/4)}{D + \beta} - \frac{(1 - d - \beta/2)d^2(3D - d)}{2D^2} + \frac{1 - d}{2(D + 1 - d)}. \tag{A34}$$

Holding (d, D) fixed,

$$\frac{\partial T}{\partial \beta} = \frac{D^2 + D\beta/2 + \beta^2/4}{(D + \beta)^2} + \frac{d^2(3D - d)}{4D^2} > 0. \tag{A35}$$

It is therefore enough to show $T(0, d, D) > 0$. Let $\eta := d/D \in (0, 1)$. Then

$$T(0, d, D) = \frac{\eta}{4[d + \eta(1 - d)]} N(d, \eta), \tag{A36}$$

where

$$N(d, \eta) = 2d(1 - d)^2\eta^2 + 3d(1 - d)(3d - 2)\eta + 7d^3 - 6d^2 - 2d + 2. \tag{A37}$$

It remains only to verify that this quadratic is positive over $(d, \eta) \in (0, 1)^2$. Its vertex is

$$\eta^*(d) = \frac{3(2 - 3d)}{4(1 - d)}.$$

If $d \le 2/5$, then $\eta^*(d) \ge 1$, so the minimum over $\eta \in [0, 1]$ occurs at $\eta = 1$, where $N(d, 1) = 5d^2 - 6d + 2 > 0$. If $2/5 < d < 2/3$, the vertex is interior and

$$N(d, \eta^*) = -\frac{(5d - 4)(5d^2 - 8d + 4)}{8} > 0.$$

Finally, if $d \ge 2/3$, then $\eta^*(d) \le 0$, so the minimum occurs at $\eta = 0$. In this case let $g(d) := N(d, 0) = 7d^3 - 6d^2 - 2d + 2$. The only critical point of g in $[2/3, 1]$ is

$$d^* = \frac{6 + \sqrt{78}}{21},$$

where

$$g(d^*) = \frac{486 - 52\sqrt{78}}{441} > 0.$$

Thus $N(d, \eta) > 0$ throughout the relevant region, and (A35) implies $Q_2 > 0$.

All three coefficients in (A18) are strictly positive, completing the verification of part (ii) of Lemma A2.

C Proofs for repression extension

This section supplies proofs and supplementary results for the repression extension.

For completeness, the regime minimizes the movement's eventual probability of success plus the expected costs of repression and violent contention. Thus, for any policy profile π, write its expected loss as

$$J(\pi) = \Pr_{\pi}(\text{movement succeeds}) + E_{\pi}\left[\sum_{t} \kappa_{\tau_t}(1-\gamma_t^{\tau_t})\right] + \lambda\, \mathbb{E}_{\pi}[N_v],$$

where N_v is the number of periods in which violence is used. The results below first characterize how repression changes the movement's success probability; repression costs and λ determine which of these interventions the regime optimally chooses.

If tactic τ is repressed in period 2 by factor $\gamma \in [0,1]$, its participation cutoff is

$$\hat{c}^{\tau}_{2,\gamma}(x) = \int_0^1 F_x(\gamma A_\tau(\ell))\, d\ell.$$

Let $L(c,u)$ denote turnout generated by participation cutoff c and common participation-cost realization u. By Lemma 2, $L(c,u)$ is weakly increasing in c for every u. Define the movement's actual period-2 success probability under the repressed tactic by

$$S^{\tau}_{2,\gamma}(x) = \mathbb{E}_u\Big[F_x\Big(\gamma A_\tau\Big(L(\hat{c}^{\tau}_{2,\gamma}(x),u)\Big)\Big)\Big],$$

and write $S^{\tau}_2(x) := S^{\tau}_{2,1}(x)$.

C.1 Responsive repression

Lemma A3 (Success under tactic-specific repression)**.** *Fix x and τ. The function $S^{\tau}_{2,\gamma}(x)$ is continuous and weakly increasing in γ. It is strictly increasing over any interval on which it is positive.*

Proof. Take $\gamma_2 > \gamma_1$. Since F_x is weakly increasing,

$$\hat{c}^{\tau}_{2,\gamma_2}(x) \geq \hat{c}^{\tau}_{2,\gamma_1}(x).$$

Couple the two repression levels using the same realization u of the common participation-cost shock. Monotonicity of turnout in the participation cutoff gives

$$L\Big(\hat{c}^{\tau}_{2,\gamma_2}(x),u\Big) \geq L\Big(\hat{c}^{\tau}_{2,\gamma_1}(x),u\Big).$$

Because A_τ is weakly increasing,

$$\begin{aligned}\gamma_2 A_\tau\Big(L(\hat{c}^{\tau}_{2,\gamma_2}(x),u)\Big) &\geq \gamma_2 A_\tau\Big(L(\hat{c}^{\tau}_{2,\gamma_1}(x),u)\Big)\\ &\geq \gamma_1 A_\tau\Big(L(\hat{c}^{\tau}_{2,\gamma_1}(x),u)\Big).\end{aligned}$$

Applying the increasing function F_x and averaging over u establishes weak monotonicity. If the lower-repression-level success probability is positive, a positive-measure set of common shocks generates power strictly above the posterior lower bound. On this set, increasing γ strictly increases realized power, and the posterior has positive density over the relevant interval, yielding strict monotonicity. Continuity follows from continuity of the cutoff and realized-power mappings together with dominated convergence. □

The next result is the formal version of the substitution cap stated verbally in Proposition 5.

Proposition A2 (Formal substitution cap)**.** *Fix a belief state x and tactics $\tau \neq \tau'$ satisfying*

$$S_2^{\tau}(x) > S_2^{\tau'}(x) > 0.$$

Suppose only tactic τ is repressed. Then there exists a unique threshold

$$\hat{\gamma}_{\tau}(x) \in (0, 1)$$

such that the organizer prefers τ for $\gamma > \hat{\gamma}_{\tau}(x)$ and τ' for $\gamma < \hat{\gamma}_{\tau}(x)$. Defining

$$V_2(x, \gamma) = \max\left\{S_{2,\gamma}^{\tau}(x), S_2^{\tau'}(x)\right\},$$

we have

$$V_2(x, 1) - V_2(x, \gamma) \leq S_2^{\tau}(x) - S_2^{\tau'}(x)$$

for every $\gamma \in [0, 1]$, with equality for every $\gamma \leq \hat{\gamma}_{\tau}(x)$.

Proof. By Lemma A3, $S_{2,\gamma}^{\tau}(x)$ is continuous and strictly increasing wherever it takes the positive fallback value. At $\gamma = 1$,

$$S_{2,1}^{\tau}(x) > S_2^{\tau'}(x),$$

whereas at $\gamma = 0$,

$$S_{2,0}^{\tau}(x) = 0 < S_2^{\tau'}(x).$$

The intermediate value theorem therefore gives a unique $\hat{\gamma}_{\tau}(x)$ satisfying

$$S_{2,\hat{\gamma}_{\tau}(x)}^{\tau}(x) = S_2^{\tau'}(x).$$

Hence

$$V_2(x, \gamma) = \begin{cases} S_{2,\gamma}^{\tau}(x), & \gamma > \hat{\gamma}_{\tau}(x), \\ S_2^{\tau'}(x), & \gamma \leq \hat{\gamma}_{\tau}(x), \end{cases}$$

up to tie-breaking at equality, and the stated bound follows immediately. Proposition 5 is the verbal implication of this result. □

The violence-aversion parameter λ generates an additional asymmetry at the substitution boundary.

Corollary A1 (Regime incentives at the substitution boundary)**.** *Suppose repression costs are strictly increasing in intensity and the regime can target only one tactic in period 2.*

(i) If peace is initially preferred and violence is the viable fallback, no optimal crowd-control policy strictly overshoots the switching threshold to violence.

(ii) If violence is initially preferred and peace is the viable fallback, then for sufficiently large λ there exists a counterinsurgency policy that induces peace and strictly dominates no repression.

Proof. For part (i), let $\hat{\gamma}_p(x)$ be the switching threshold. For every $\gamma < \hat{\gamma}_p(x)$, the organizer has already abandoned peace, so its success probability is fixed at the unrepressed violent fallback value. Replacing γ by $\hat{\gamma}_p(x)$ weakly preserves that success probability while strictly lowering repression costs; depending on tie-breaking, it also weakly lowers the violence cost. Thus every strict overshoot is dominated.

For part (ii), choose any $\gamma < \hat{\gamma}_v(x)$ that induces peace. Relative to no repression, the induced-peace policy replaces the initially preferred violent success probability $S_2^v(x)$ with the lower fallback value $S_2^p(x)$, incurs finite repression cost $\kappa_v(1-\gamma)$, and avoids the direct violence cost λ. Hence it strictly dominates no repression whenever

$$\lambda > \kappa_v(1-\gamma) - \Big(S_2^v(x) - S_2^p(x)\Big).$$

The right-hand side is finite, so the claim follows. □

C.2 Preemptive repression and the belief refund

The main-text Proposition 6 suppresses two auxiliary objects. For a fixed continuation tactic τ, define its unconditional contribution to lifetime success after a first-period failure at power x by

$$C_\rho^\tau(x) := \frac{M-x}{M} S_2^\tau(x).$$

Let

$$C_\rho(x) := \max_{\tau \in \{p,v\}} C_\rho^\tau(x)$$

and define the movement's eventual success probability conditional on first-period realized power x by

$$H_\rho(x) = \frac{x}{M} + C_\rho(x).$$

Proposition A3 (Formal belief-refund decomposition)**.** *For every $x' < x$,*

$$C_\rho(x') \geq C_\rho(x),$$

and therefore

$$H_\rho(x) - H_\rho(x') = \frac{x - x'}{M} - \left[C_\rho(x') - C_\rho(x)\right].$$

Thus lowering realized first-period power weakly improves continuation prospects and offsets part of the direct loss in first-period success. Moreover, for

$$0 \le x' < x \le \min\{\alpha, c_\rho(\alpha, \beta; M)\},$$

we have

$$H_\rho(x') = H_\rho(x) \quad \textit{if } \rho\beta = 0,$$

while

$$H_\rho(x') > H_\rho(x) \quad \textit{if } \rho > 0 \textit{ and } \beta > 0.$$

Proof. Fix a continuation tactic τ and consider $x' < x$. Under the lower posterior truncation point x', citizens are weakly more optimistic, so

$$\hat{c}_2^\tau(x') \ge \hat{c}_2^\tau(x).$$

Using the same common shock u in both continuation games gives

$$L(\hat{c}_2^\tau(x'), u) \ge L(\hat{c}_2^\tau(x), u),$$

and hence

$$A_\tau(L(\hat{c}_2^\tau(x'), u)) \ge A_\tau(L(\hat{c}_2^\tau(x), u)).$$

Conditional on u, the continuation-success interval after failure at x is therefore contained in the corresponding interval after x'. Averaging over u yields

$$C_\rho^\tau(x') \ge C_\rho^\tau(x).$$

Taking the maximum over tactics preserves the inequality, establishing the decomposition.

For the full- and over-refund claims, let

$$z \le \min\{\alpha, c_\rho(\alpha, \beta; M)\}.$$

Because $z \le c_\rho$, violence is optimal in continuation. Because $z \le \alpha$, even zero-turnout violence reaches the posterior lower bound. Its participation cutoff is therefore

$$\hat{c}_2^v(z) = \frac{\alpha + \beta/2 - z}{M - z}.$$

Using the turnout distribution from Lemma 2, the actual period-2 success probability is

$$S_2^v(z) = \rho\left(1 - \hat{c}_2^v(z)\right)\frac{\alpha - z}{M - z} + (1 - \rho)\hat{c}_2^v(z) + \rho\hat{c}_2^v(z)\frac{\alpha + \beta - z}{M - z}.$$

Substituting into lifetime success gives

$$H_\rho(z) = \frac{\alpha + (1-\rho)\beta/2}{M} + \frac{\rho\beta(\alpha + \beta/2 - z)}{M(M-z)}.$$

Differentiating,

$$H'_\rho(z) = -\frac{\rho\beta\,(M - \alpha - \beta/2)}{M(M-z)^2}.$$

Since $M > 1 > \alpha + \beta$, the final factor is strictly positive. Thus $H'_\rho(z) = 0$ when $\rho\beta = 0$ and $H'_\rho(z) < 0$ when $\rho > 0$ and $\beta > 0$, proving the result. Proposition 6 follows immediately. □

C.3 Persistence of violence

The no-period-2-repression assumption in Proposition 7 is a benchmark used to isolate the informational channel. Allowing the regime to re-optimize repression after the first failure would superimpose the responsive-substitution mechanism characterized above: in particular, second-period counterinsurgency could itself prevent the violent recurrence whose informational source we want to identify here.

Let γ scale only the period-1 violent technology. Under the prior, the period-1 participation cutoff when the organizer opens violently is

$$\hat{c}^v_{1,\gamma} = \int_0^1 \frac{\gamma(\alpha + \beta\ell)}{M}\, d\ell = \frac{\gamma(\alpha + \beta/2)}{M}.$$

Let X_γ denote realized first-period violent power under this policy, and let $c_\rho = c_\rho(\alpha, \beta; M)$ denote the continuation cutoff from Proposition 3.

Lemma A4 (Nested violent-recurrence events)**.** *Consider two counterinsurgency policies $\gamma' < \gamma$ under which the organizer opens violently in both cases. Coupling them using the same common participation-cost shock and regime type gives*

$$X_{\gamma'} < X_\gamma.$$

Consequently, if violence recurs in period 2 under γ, then it also recurs under the stronger policy γ'.

Proof. Because $\gamma' < \gamma$,

$$\hat{c}^v_{1,\gamma'} < \hat{c}^v_{1,\gamma}.$$

Under the same common participation-cost shock,

$$\ell_{\gamma'} \le \ell_\gamma,$$

so

$$X_{\gamma'} = \gamma'(\alpha + \beta\ell_{\gamma'}) < \gamma(\alpha + \beta\ell_\gamma) = X_\gamma.$$

Violence recurs under γ exactly when

$$\theta \geq X_\gamma \qquad \text{and} \qquad X_\gamma < c_\rho.$$

These inequalities imply

$$\theta \geq X_\gamma > X_{\gamma'} \qquad \text{and} \qquad X_{\gamma'} < X_\gamma < c_\rho,$$

which are sufficient for recurrence under γ'. □

Proof of Proposition 7. By Lemma A4, the event of a second violent episode is weakly larger under stronger counterinsurgency as long as both policies preserve a violent opening. Since violence occurs with certainty in period 1 on both branches, the expected number of violent periods is therefore weakly larger under the stronger policy.

If counterinsurgency instead induces a peaceful opening, violence can occur only in period 2. Its expected incidence is strictly below one because peaceful protest has a strictly positive probability of succeeding in period 1. Every policy that preserves a violent opening, by contrast, generates one violent period with certainty and therefore has expected violent incidence at least one. Any policy inducing a peaceful opening consequently produces strictly less expected violence. □